Testing Evidence

MMM Social Workers & Associates — Booking System Application

Module	ITDMA3-34 — Project: Mobile Application and Web Services (Eduvos)
Group	Group 11
Group members	Faith Murau (EDUV4821116) | Khauhelo Moqechane (MD.2022.Q7M4P1) | Ignatius Mayibongwe Khumalo (EDUV4960805)
Purpose	Screenshot and recording evidence for every test ID recorded in TESTING.md
App version	1.0 (Android, Java; Firebase backend, region africa-south1)
Test period	From 15/10/2026 to 30/10/2026

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Results at a glance

Group	Total	PASS	PENDING
Automated unit tests	61	61	0
Hardening verification (PA1 to PF9)	37	37	0
Functional plan (F1 to F25)	25	25	0
Usability (U1 to U12)	12	12	0
Edge cases (E1 to E8)	8	8	0
Team: UX tests (UX-01 to UX-24)	24	24	0
Team: Navigation tests (NAV-01 to NAV-18)	18	18	0
Team: Database tests (DB-01 to DB-21)	21	21	0
Team: Booking logic tests (BK-01 to BK-22)	22	22	0
Team: Permission tests (PM-01 to PM-18)	18	18	0
Team: Integration tests (INT-01 to INT-12)	12	12	0

1. Automated tests

61 unit tests: ValidationUtilsTest (37) and BookingRulesTest (24). Run: right-click app/src/test/java > Run Tests.

UT1 — All unit tests pass

Test	Run every unit test in app/src/test/java
Expected result	61 tests, all green
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

The Android Studio test results window showing 61 passed

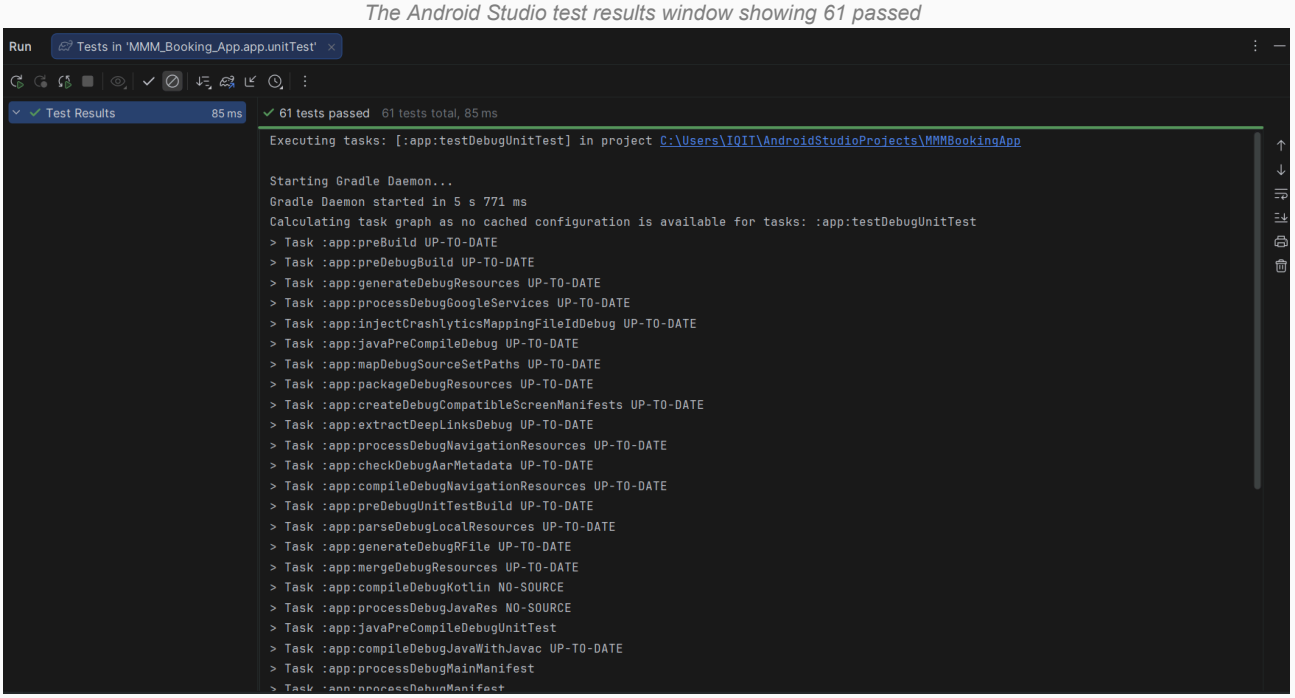
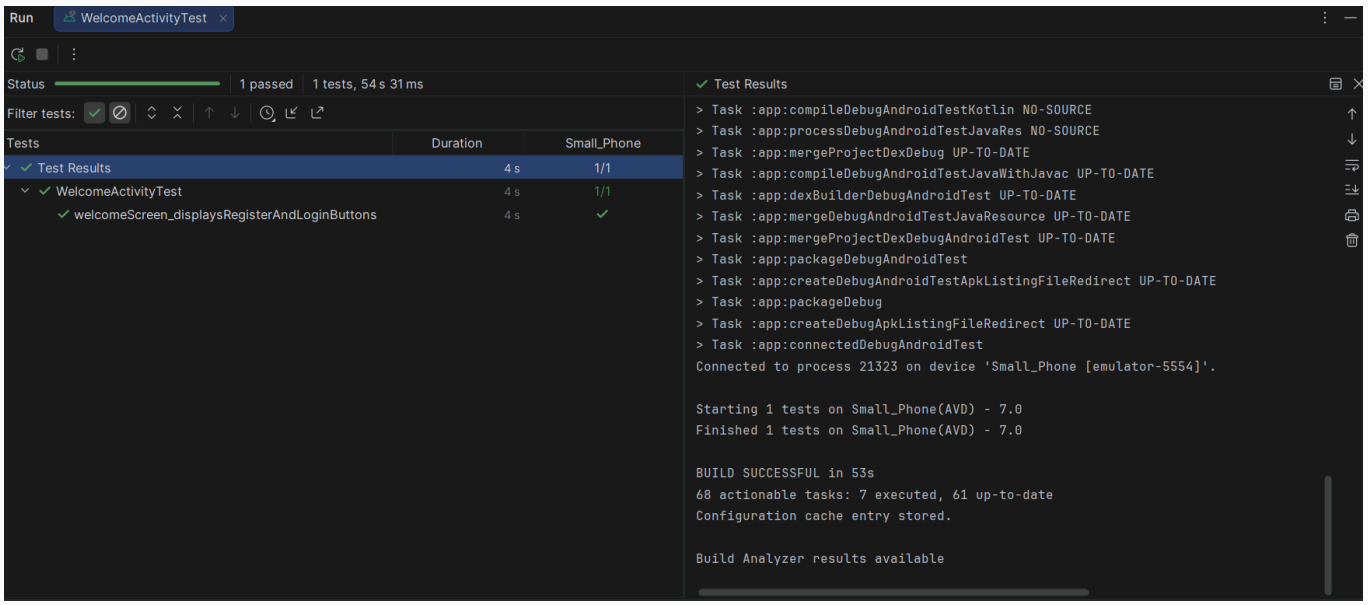


Figure UT1

UT2 — Espresso test on the Welcome screen

Test	Run WelcomeActivityTest on the connected device
Expected result	Register and Login buttons are visible and correctly labelled; test passes
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026



The test result

Figure UT2

2. Hardening verification tests (PA1 to PF9)

Security, data-integrity and robustness work from the final implementation audit. Each was run on the real phone and passed.

2.1 Phase A — Security rules

PA1 — Register a brand-new test account

Expected result	Account created normally; role defaults to client
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

Registration success dialog

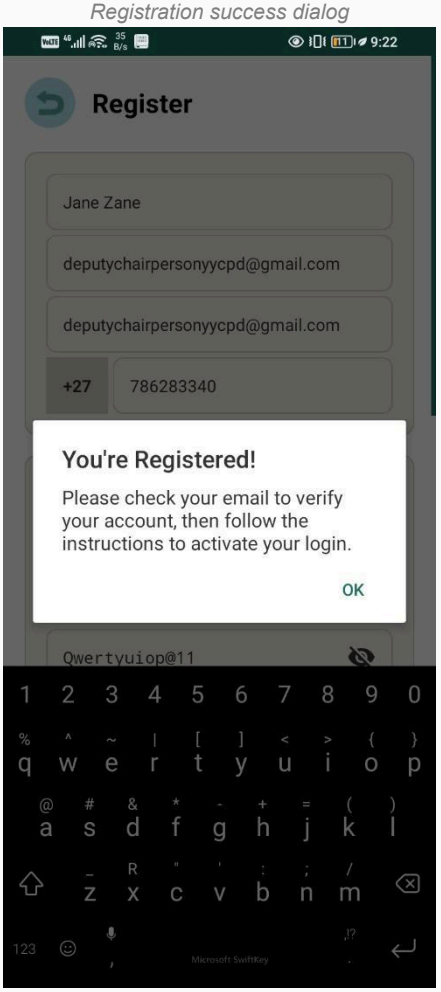


Figure PA1

PA2 — Confirm a client cannot set status/paid on a booking (Firebase Rules Playground)

Expected result	The write is denied by the rules
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

Rules Playground screenshot showing the write denied

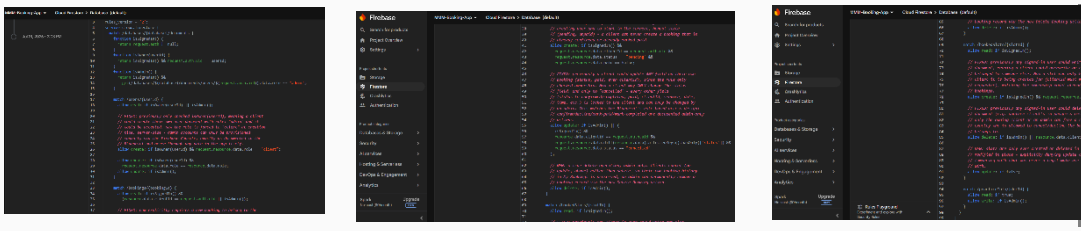


Figure PA2

PA3—Register, book and cancel as a client

Expected result	Whole flow still works under the new rules
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

Booking confirmation, then the booking showing Cancelled in My Bookings

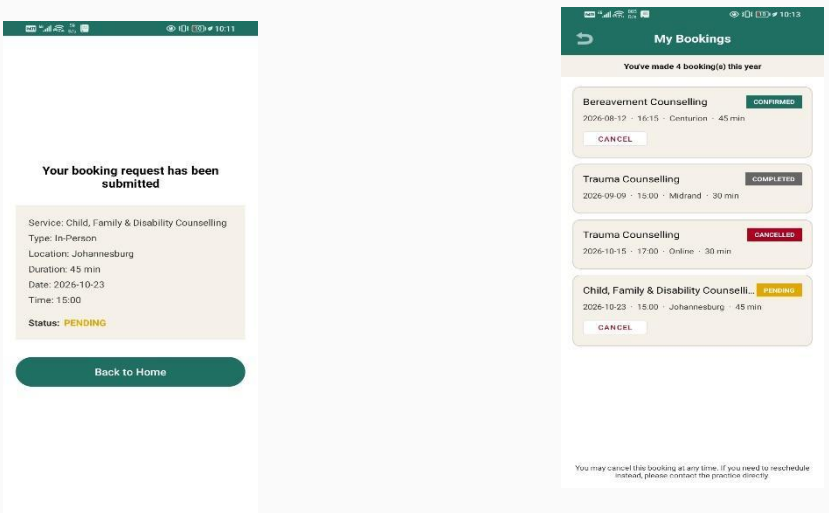


Figure PA3

2.2 Phase B — Booking CRUD (atomic cancel, true delete)

PB1—Client cancels a pending booking

Expected result	Shows Cancelled; the same time slot can be booked again

Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

Cancelled status in My Bookings, then the same slot booked again

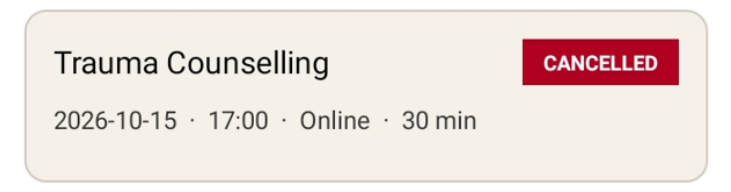


Figure PB1

PB2—Admin declines a pending booking

Expected result	Confirmation dialog first; then Cancelled and slot freed
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

Decline confirmation dialog, then the booking showing Cancelled

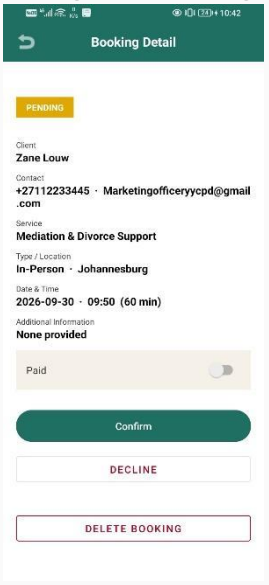


Figure PB2

PB3—Admin deletes a booking

Expected result	Confirmation first; screen closes; booking disappears from the dashboard; the bookings document is gone in the Firebase Console
Result	**PASS**

Device used	Phone and Table Date: 15/10/2026

Delete dialog, dashboard without the booking, and the Firebase bookings collection without that document

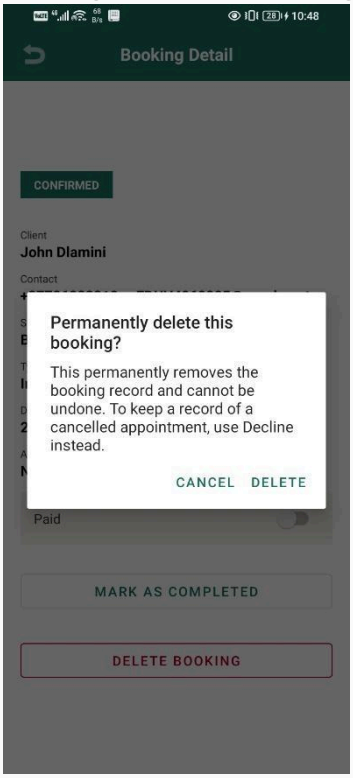


Figure PB3

PB4—Admin deletes a cancelled booking whose slot was since re-booked by another client

Expected result	The other client's slot document still exists
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

Firebase bookedSlots showing the other client's slot still present after the delete

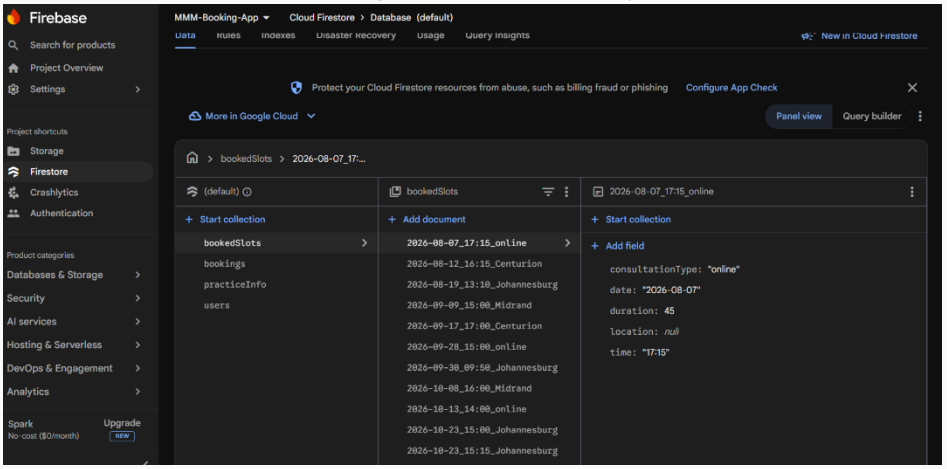


Figure PB4

PB5—Client has no delete action

Expected result	No client delete button; the server-side rule is the protection
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

Client screens with no delete option



Figure PB5

2.3 Phase C — Booking rules

PC1 — Closing time (hours 08:00–18:00)

Expected result	60 min at 17:45 rejected; 17:00 accepted; 17:30 + 60 min rejected, switching to 30 min fits
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

The rejection message for 17:45 + 60 min and the accepted 17:00 booking

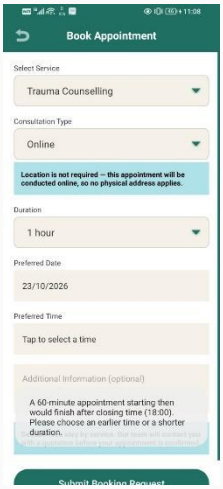


Figure PC1

PC2 — Quarter-hour starts

Expected result	10:20 rejected with the quarter-hour message; 10:15 accepted
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

The quarter-hour message for 10:20

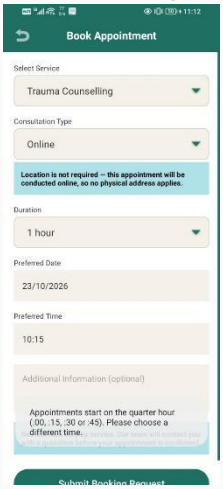


Figure PC2

PC3 — Cross-client overlap

Expected result	Client A books 10:00 (60 min); Client B at 10:30 and at 10:00 are both rejected; 11:00 is accepted
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

Client B's rejection message (both attempts) and the accepted 11:00 booking

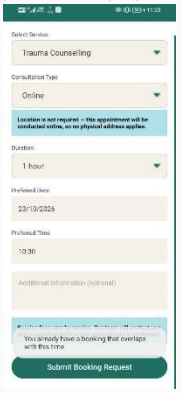


Figure PC3

PC4 — Same-client overlap

Expected result	Client A booking 10:30 while holding 10:00–11:00 gets the 'you already have a booking that overlaps' message
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

The own-overlap message

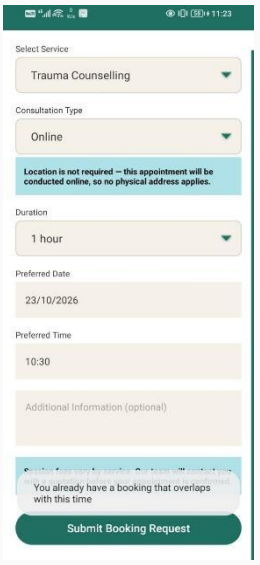


Figure PC4

PC5 — Slot freed on cancel

Expected result	After A cancels, B can book 10:00 straight away

Result	PASS
Device used	Phone and Table Date: 15/10/2026

Client B's successful booking at 10:00 after A's cancellation

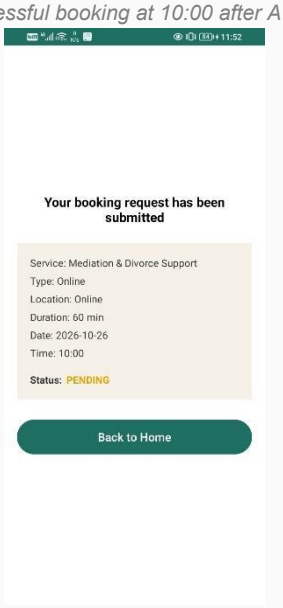
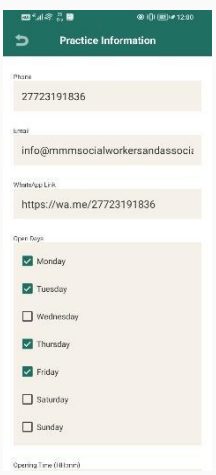


Figure PC5

PC6 — Operating days

Expected result	Untick Wednesday as admin; a client cannot pick a Wednesday; Contact shows 'Monday, Tuesday, Thursday - Friday'; restore afterwards
Result	PASS
Device used	Phone and Table Date: 15/10/2026

Practice Information with Wednesday unticked, the closed-day message, and the Contact screen text.



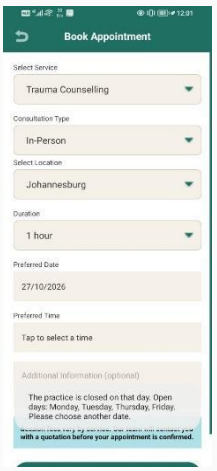


Figure PC6

PC7 — Admin hours validation

Expected result	8:00, 17:00 and opening-after-closing each rejected; saving with no days ticked rejected
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

Each validation message

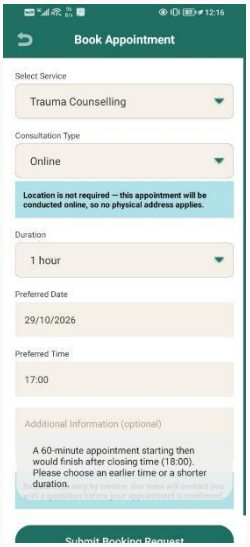


Figure PC7

PC8 — Future-year holiday

Expected result	A date in next January is checked without error; 1 January is rejected
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

The holiday-rejection message for 1 January

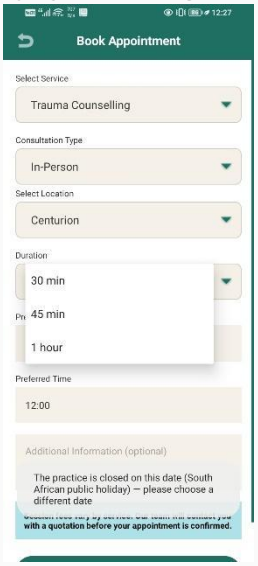


Figure PC8

2.4 Phase D — Booking Reminders switch

PD1 — Switch ON (default): book an appointment

Expected result	'Booking Submitted' notification appears
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

The Booking Submitted notification

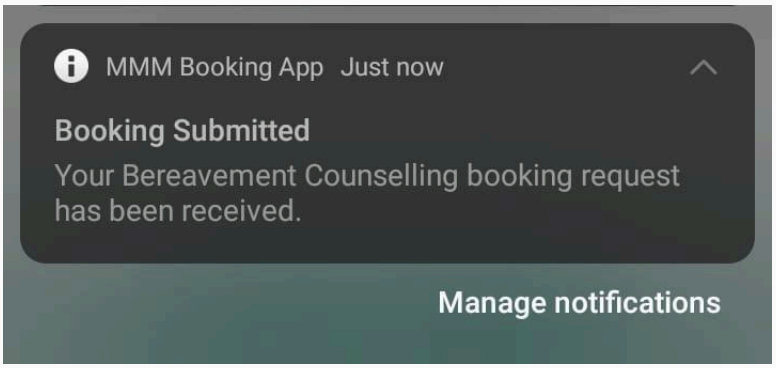


Figure PD1

PD2 — Switch OFF, then book

Expected result	'Disabled' toast; no notification; booking still succeeds
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

The disabled toast and the absence of a notification after booking

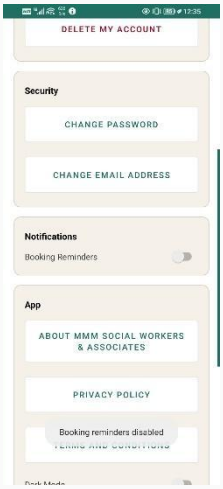


Figure PD2

PD3 — Re-open Settings several times

Expected result	No toast on opening — a toast only appears when the switch is tapped
Result	**PASS**

Device used	Phone and Table Date: 15/10/2026

Settings opened with no toast)

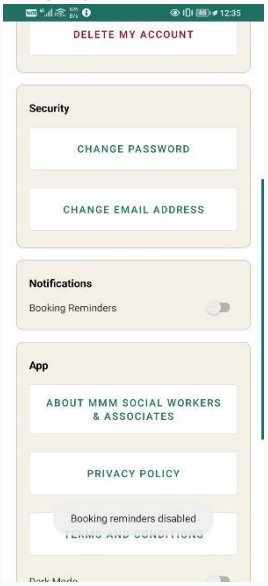


Figure PD3

PD4 — Scheduled reminder cancelled when OFF, restored when ON (adb shell dumpsys alarm)

Expected result	Alarm present when ON, gone when OFF, back when ON again
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

The dumpsys output for ON, OFF and ON again

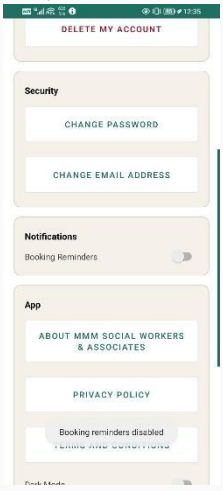


Figure PD4

PD5 — Persistence

Expected result	Still OFF after fully closing the app; notificationsEnabled is false in the Console
Result	**PASS**

Device used	Phone and Table Date: 15/10/2026

Settings showing OFF after reopening, and the Console field

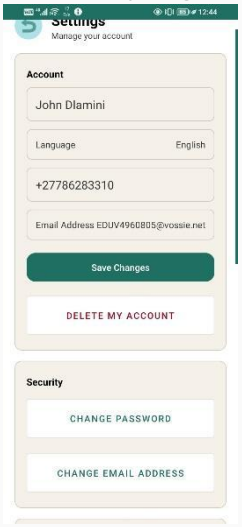


Figure PD5

PD6 — Account isolation

Expected result	A second account on the same phone shows the switch ON
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

The second account's Settings with the switch ON

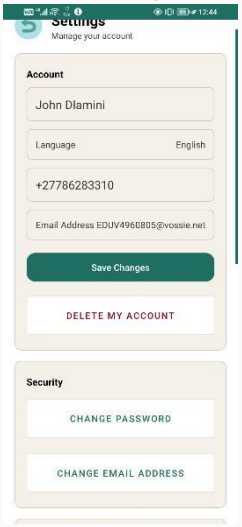


Figure PD6

PD7 — Dark Mode toggle

Expected result	Still works as before
Result	**PASS**

Device used	Phone and Table Date: 15/10/2026

Dark Mode on and off

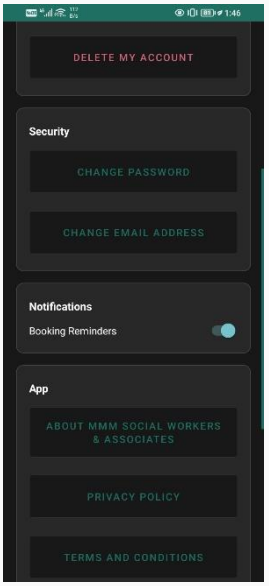


Figure PD7

2.5 Phase E — Robustness

PE1 — Registration rollback (profile creation temporarily blocked in the rules)

Expected result	'Registration failed' toast; the new email is not left in Authentication > Users
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

The failure toast and the Authentication Users list without the new email



Figure PE1

PE2 — Delete a booking's status field in the Console

Expected result	Admin Dashboard does not crash; booking shown/counted as Pending; Booking Detail opens

Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

The dashboard with the booking counted as Pending, and its Booking Detail

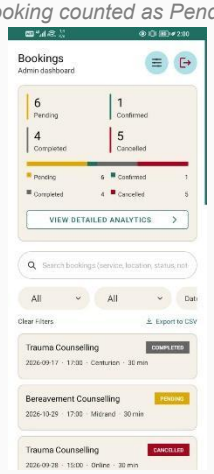


Figure PE2

PE3 — Reopen the app in Airplane Mode (client and admin)

Expected result	Lands on Home / Admin Dashboard, not Welcome
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

Home and Admin Dashboard opened in Airplane Mode

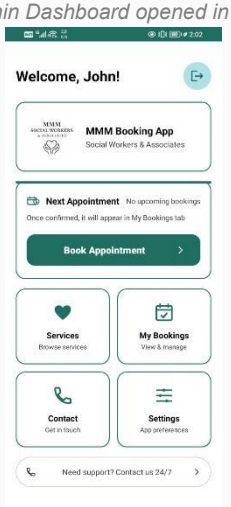


Figure PE3

PE4 — Regression: login routing and back button

Expected result	Client to Home, admin to Dashboard with the 'Logged in successfully' toast; double-back-to-exit works
Result	**PASS**

Device used	Phone and Table Date: 15/10/2026

The success toast on each home screen, and the back-press toast

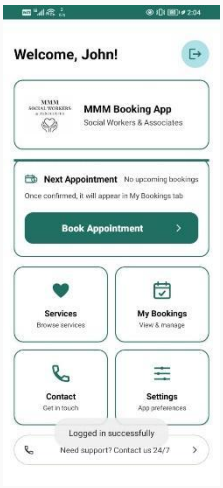


Figure PE4

PE5 — Logcat check

Expected result	No email address in any LoginActivity/RegisterActivity log line
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

Logcat filtered to LoginActivity and RegisterActivity



Figure PE5

2.6 Phase F — Clean-up and validation

PF1 — Clean build after deleting the default test stubs

Expected result	Compiles with no errors
Result	**PASS**

Device used	Phone and Table Date: 15/10/2026

Successful build output

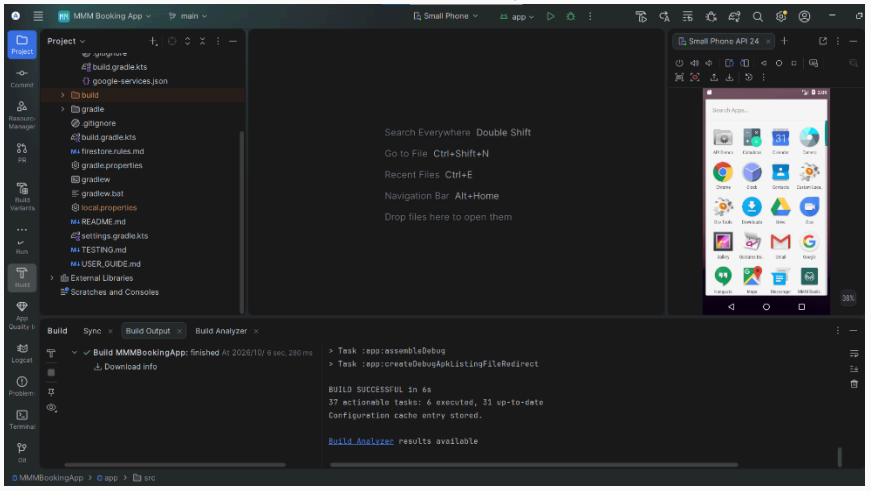


Figure PF1

PF2 — Run all unit tests

Expected result	61 tests, all green
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

The Android Studio test results window showing 61 passed

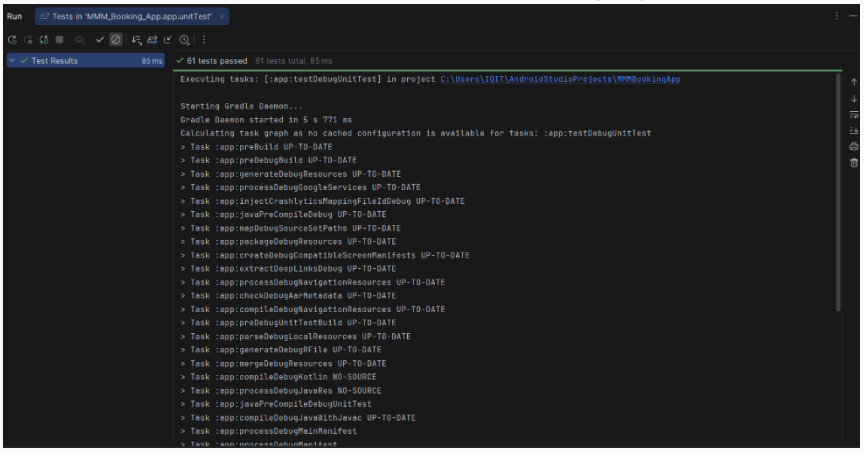


Figure PF2

PF3 — Register with email 'thandi@example'

Expected result	Email error shown; thandi@example.com accepted
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

The email error for thandi@example

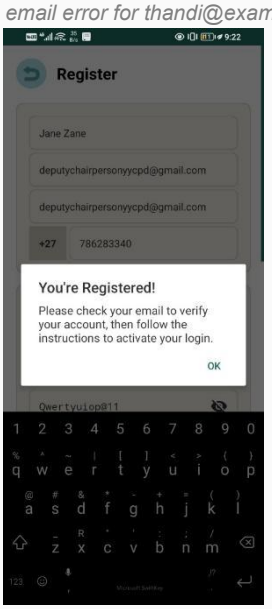


Figure PF3

PF4 — Register with a weak password

Expected result	Rules line turns red and bold, no toast; normal when typing, green when valid
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

The red, bold password rules line after submit



Figure PF4

PF5 — Log In with empty fields

Expected result	'This field is required' on each empty field
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

The required-field messages on Login

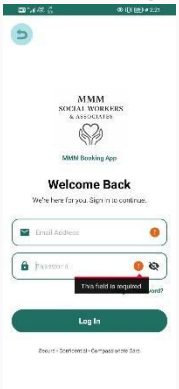
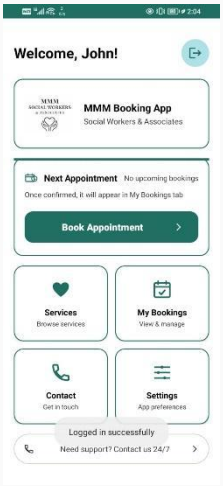


Figure PF5

PF6 — Normal login

Expected result	Client to Home, admin to Dashboard with the success toast
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

Home and Admin Dashboard after login



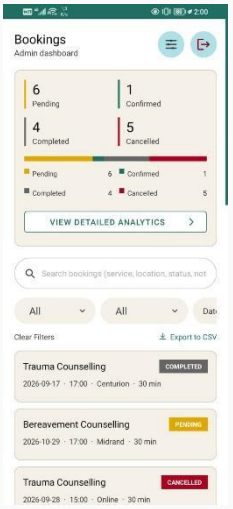


Figure PF6

PF7 — Booking Detail as admin

Expected result	Badge colour and Confirm/Decline/Complete/Delete behave as before; confirmation dialogs appear
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

Booking Detail as admin

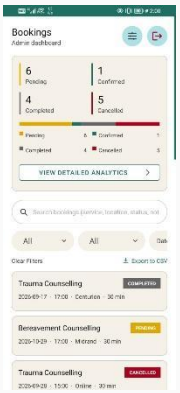


Figure PF7

PF8 — Comment-only files (Home, Change Email, Admin Analytics)

Expected result	Behave exactly as before
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

Home, Change Email and Admin Analytics screens



Figure PF8

PF9 — git status after untracking .idea

Expected result	.idea no longer tracked
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

Terminal output of git status

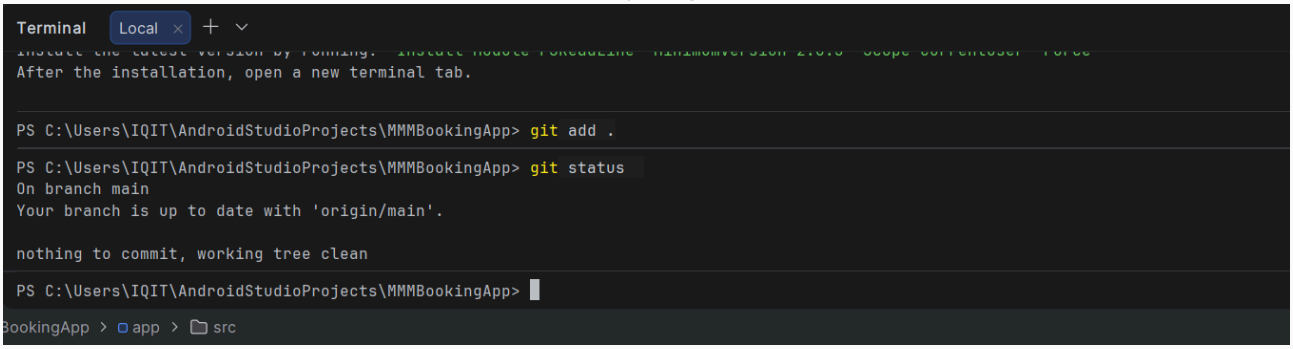


Figure PF9

3. Functional test plan (F1 to F25)

F1 — Registration

Steps	Fill all fields correctly, tick consent, submit
Expected result	Account created, verification email sent
Result	**PASS (PA1)**
Device used	Phone and Table Date: 15/10/2026

A screenshot of the result described above

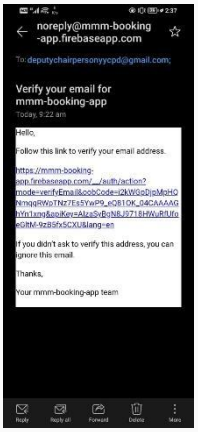


Figure F1

F2 — Registration blocked without consent

Steps	Fill all fields, leave consent unticked, submit
Expected result	Red error shown, no account created
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

A screenshot of the result described above

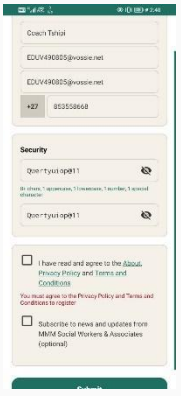


Figure F2

F3 — Email verification gate

Steps	Log in before verifying email
Expected result	Blocked, resend option shown
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

A screenshot of the result described above

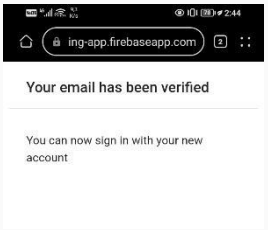


Figure F3

F4 — Login (verified account)

Steps	Log in with correct credentials
Expected result	Routed to Home (client) or Admin Dashboard (admin), with a brief 'Logged in successfully' toast
Result	**PASS (PE4, PF6)**
Device used	Phone and Table Date: 15/10/2026

A screenshot of the result described above

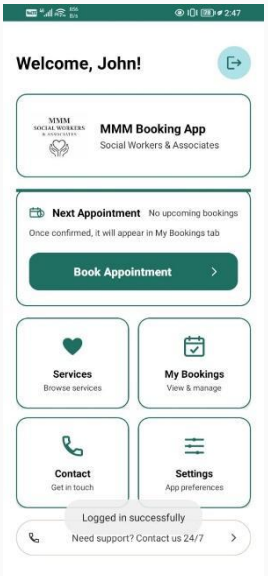


Figure F4

F5 — Password reset

Steps	Request reset for a registered email
Expected result	Reset email received
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

A screenshot of the result described above

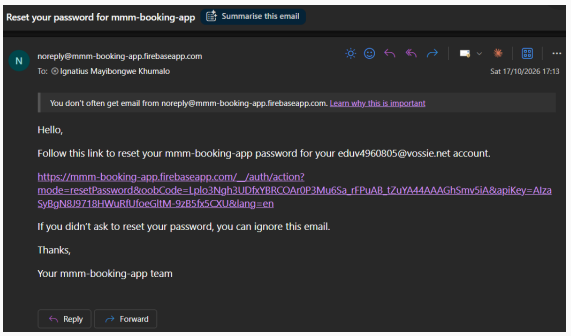


Figure F5

F6 — Booking submission

Steps	Complete and submit a valid booking
Expected result	Confirmation screen shown, booking appears in My Bookings, immediate notification received
Result	**PASS (PD1)**
Device used	Phone and Table Date: 15/10/2026

A screenshot of the result described above

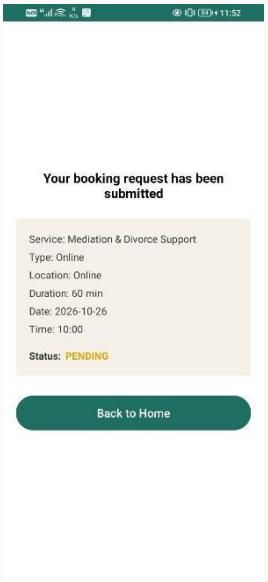


Figure F6

F7 — Double-booking prevention

Steps	Two accounts attempt overlapping times
Expected result	Second attempt rejected
Result	**PASS (PC3)**
Device used	Phone and Table Date: 15/10/2026

A screenshot of the result described above

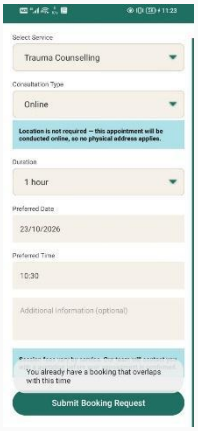


Figure F7

F8 — Public holiday block

Steps	Attempt a known SA public holiday date
Expected result	Booking date rejected with a clear message
Result	**PASS (PC8)**
Device used	Phone and Table Date: 15/10/2026

A screenshot of the result described above



Figure F8

F9 — Cancellation

Steps	Cancel a pending/confirmed booking

Expected result	Confirmation dialog shown, status becomes Cancelled, slot freed
Result	**PASS (PB1)**
Device used	Phone and Table Date: 15/10/2026

A screenshot of the result described above

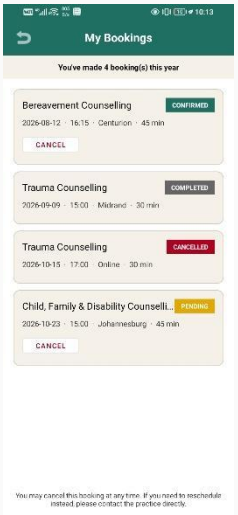


Figure F9

F10 — Admin confirm/decline

Steps	Admin confirms a pending booking
Expected result	Status updates automatically on the client's My Bookings screen in real time
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

A screenshot of the result described above

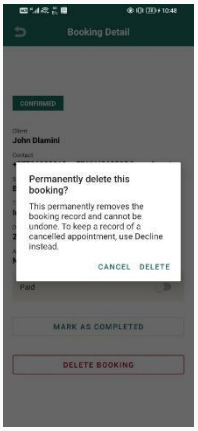


Figure F10

F11 — Admin search

Steps	Type a service name into the search box

Expected result	List filters to matching bookings only
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

A screenshot of the result described above

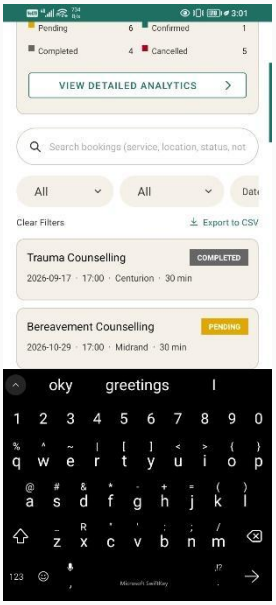


Figure F11

F12 — Admin search + filters combined

Steps	Apply a status filter, then search
Expected result	Only bookings matching both are shown
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

A screenshot of the result described above

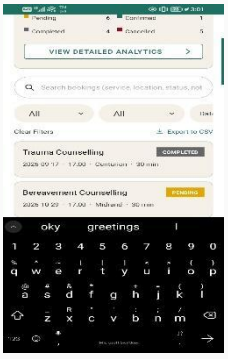


Figure F12

F13 — CSV export

Steps	Tap Export CSV on Admin Dashboard
Expected result	Share sheet opens with a valid CSV file
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

A screenshot of the result described above

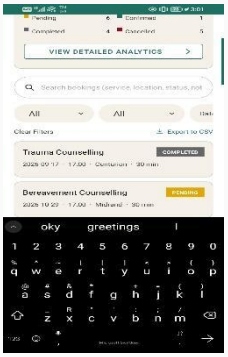


Figure F13

F14 — Practice Information edit

Steps	Admin changes open days/phone, saves
Expected result	Client's Contact and booking screens reflect the change
Result	**PASS (PC6)**
Device used	Phone and Table Date: 15/10/2026

A screenshot of the result described above

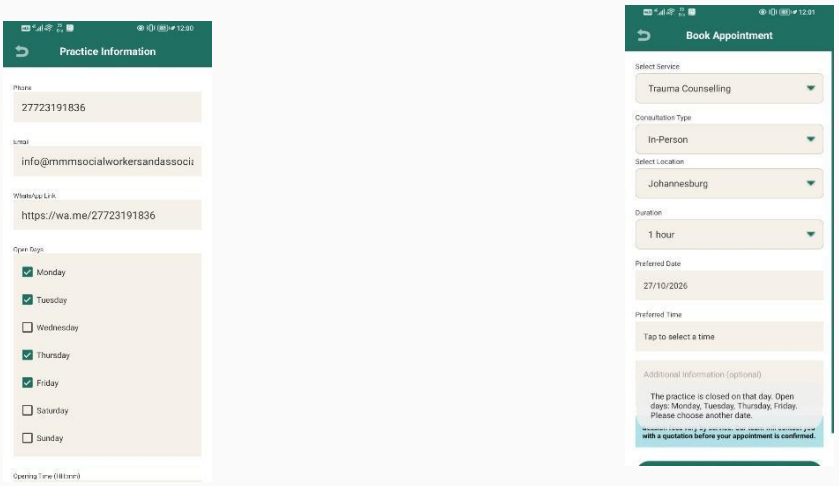


Figure F14

F15 — Admin Analytics

Steps	Open View Detailed Analytics

Expected result	Counts, % change and bar charts display for both Bookings Created and New Registrations
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

A screenshot of the result described above



Figure F15

F16 — Settings — single logout

Steps	Open Settings, scroll to Support
Expected result	Exactly one Logout button present
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

A screenshot of the result described above



Figure F16

F17 — Dark Mode toggle confirmation

Steps	Toggle Dark Mode on Settings
Expected result	Toast confirms enabled/disabled; theme changes across the app

Result	**PASS (PD7)**
Device used	Phone and Table Date: 15/10/2026

A screenshot of the result described above

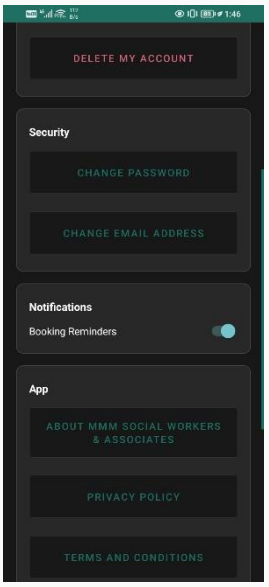


Figure F17

F18 — Booking Reminders toggle

Steps	Toggle Booking Reminders on Settings
Expected result	Toast confirms enabled/disabled; notifications obey the switch
Result	**PASS (PD2, PD3)**
Device used	Phone and Table Date: 15/10/2026

A screenshot of the result described above

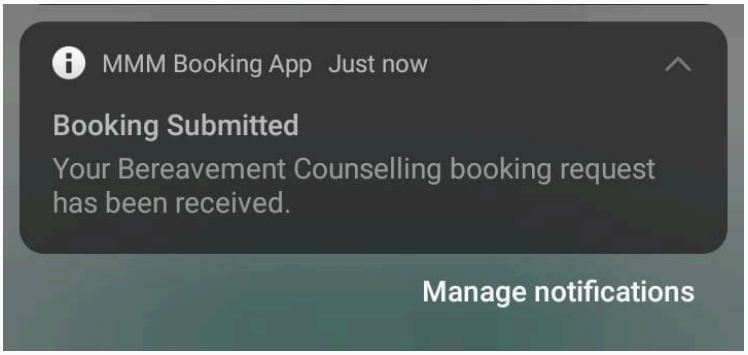


Figure F18

F19 — Logout confirmation dialog

Steps	Tap Logout from Home, Settings, and Admin Dashboard

Expected result	Cancel/Log Out dialog each time; Cancel does nothing; Log Out returns to Welcome
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

A screenshot of the result described above

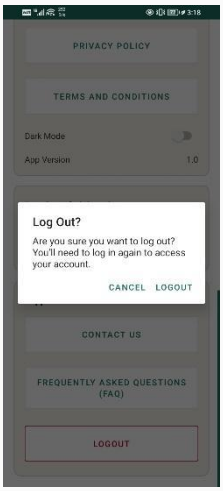


Figure F19

F20 — Delete Account explainer

Steps	Tap Delete My Account on Settings
Expected result	Explainer first; Continue opens the pre-addressed email; Cancel dismisses
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

A screenshot of the result described above

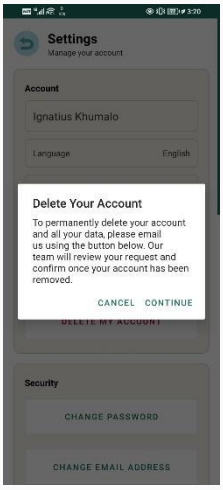


Figure F20

F21 — Back button on Home/Admin Dashboard

Steps	Press back once, then again within 2 seconds

Expected result	First press shows 'Press back again to exit'; second exits; user stays logged in
Result	**PASS (PE4)**
Device used	Phone and Table Date: 15/10/2026

A screenshot of the result described above

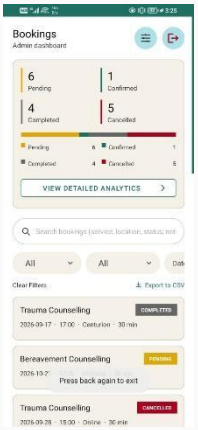


Figure F21

F22 — Session persistence

Steps	Log in, fully close the app, reopen
Expected result	Opens directly to Home/Admin Dashboard without logging in again
Result	**PASS (PE3)**
Device used	Phone and Table Date: 15/10/2026

A screenshot of the result described above

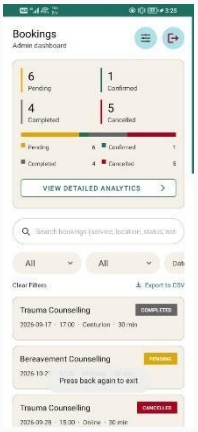


Figure F22

F23 — Real-time admin list update

Steps	With Admin Dashboard open, a client submits a booking from another device
Expected result	New booking and updated counts appear automatically
Result	**PASS**

Device used	Phone and Table Date: 15/10/2026

A screenshot of the result described above

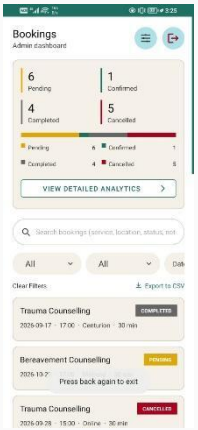


Figure F23

F24 — Confirm-email mismatch on Register

Steps	Type different addresses in Email and Confirm Email, submit
Expected result	'Email addresses do not match' shown; no account created
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

A screenshot of the result described above



Figure F24

F25 — Change Email

Steps	Settings > Change Email with current password and a new address
Expected result	'Confirmation Sent' dialog; link arrives at the new address; login email unchanged until clicked
Result	**PASS**

Device used	Phone and Table Date: 15/10/2026

A screenshot of the result described above

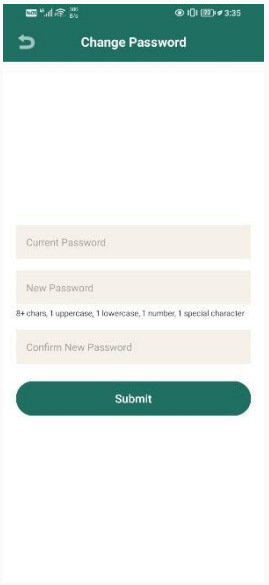


Figure F25

4. Usability tests (U1 to U12)

U1 — Dropdown fields

Expected result	Rounded border + visible arrow on all Spinners
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

A screenshot showing the point being checked (phone and/or tablet)

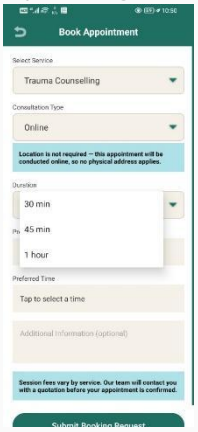


Figure U1

U2 — Password fields

Expected result	Eye icon toggles visibility correctly on Login and Register
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

A screenshot showing the point being checked (phone and/or tablet)



Figure U2

U3 — Dark Mode

Expected result	All screens switch palette, remain readable; status badges show bold white text with good contrast

Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

A screenshot showing the point being checked (phone and/or tablet)

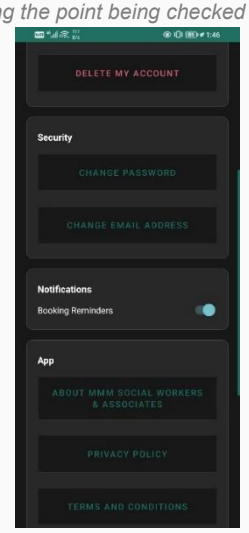


Figure U3

U4 — Scroll indicators

Expected result	Visible on screens with content exceeding one view
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

A screenshot showing the point being checked (phone and/or tablet)

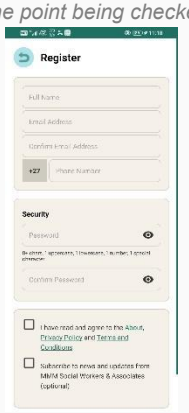


Figure U4

U5 — Touch target size

Expected result	Admin Dashboard's Logout and Settings icons comfortably tappable
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

A screenshot showing the point being checked (phone and/or tablet)

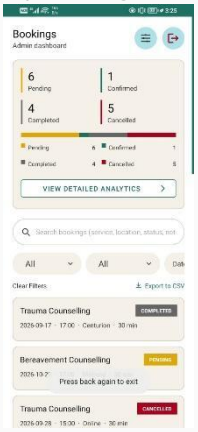


Figure U5

U6 — Register checkbox text

Expected result	Both consent sentences fully visible; '(optional)' has clear spacing below it
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

A screenshot showing the point being checked (phone and/or tablet)

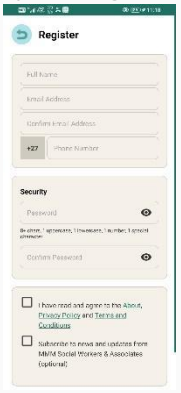


Figure U6

U7 — Settings long labels

Expected result	'About', 'FAQ' etc. wrap to two lines cleanly, never cut off
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

A screenshot showing the point being checked (phone and/or tablet)



Figure U7

U8 — Logout dialog wording

Expected result	States that logging out requires logging in again; button order Cancel (left), Log Out (right)
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

A screenshot showing the point being checked (phone and/or tablet)

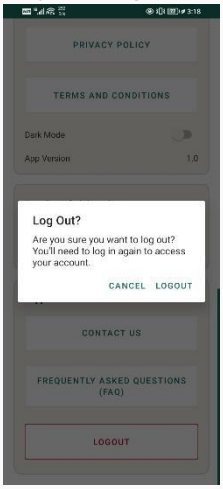


Figure U8

U9 — Tablet layout — centred content

Expected result	On a tablet, content is centred with generous margins, not stretched edge-to-edge
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

A screenshot showing the point being checked (phone and/or tablet)

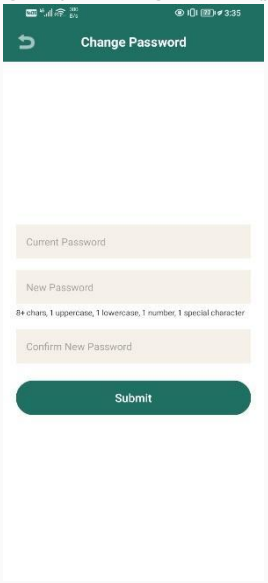


Figure U9

U10 — Tablet layout — no dead space

Expected result	Fixed-content screens balance vertically on a tall tablet screen
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

INSERT SCREENSHOT FOR U10 HERE

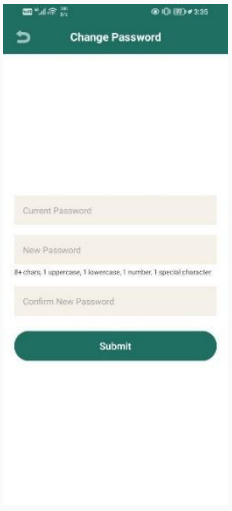


Figure U10

U11 — Tablet font scaling

Expected result	Text is visibly larger on tablet than on phone, without layout breaking
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

A screenshot showing the point being checked (phone and/or tablet)



Figure U11

U12 — Landscape orientation

Expected result	Home, Book Appointment, Register remain usable and scrollable in landscape on phone and tablet
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

A screenshot showing the point being checked (phone and/or tablet)

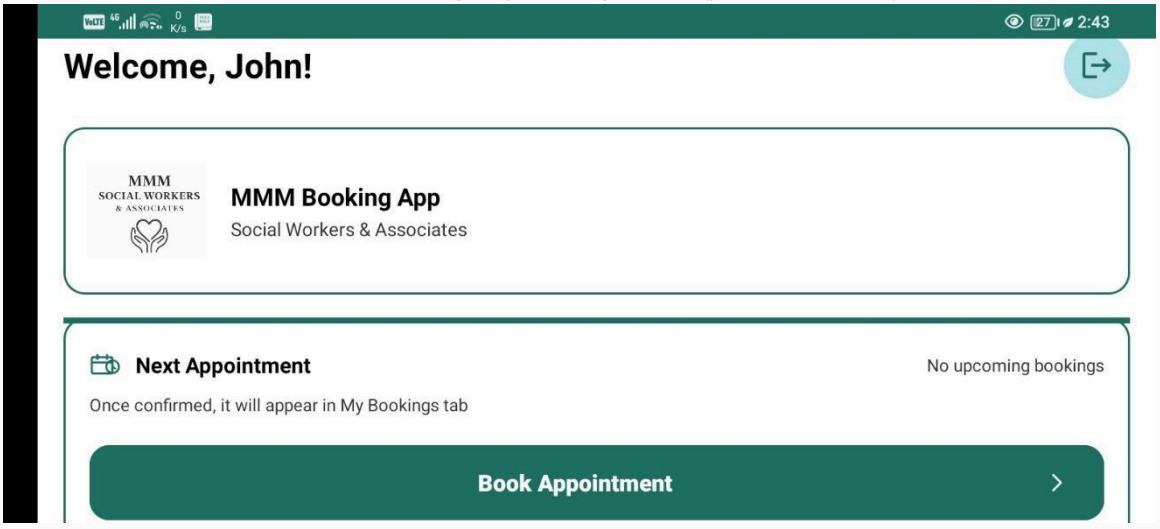


Figure U12

5. Edge-case tests (E1 to E8)

E1 — Submit booking form with fields missing

Expected result	Specific message per missing field
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

A screenshot of the message or screen shown



Figure E1

E2 — Attempt booking while offline

Expected result	Clear offline message, no crash
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

A screenshot of the message or screen shown

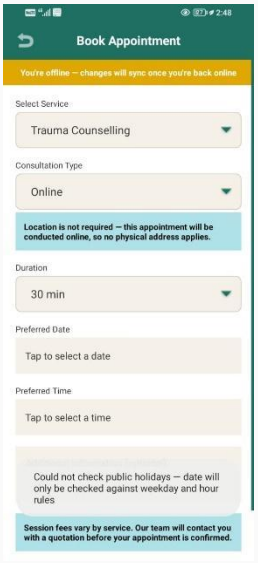


Figure E2

E3 — Attempt login while offline

Expected result	Clear offline message, no crash
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

A screenshot of the message or screen shown

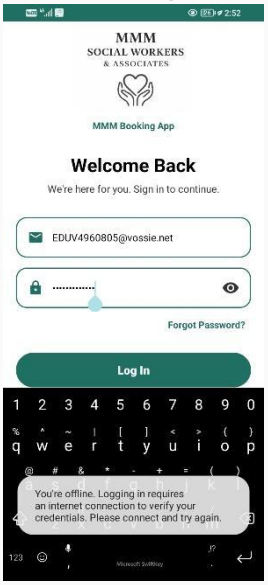


Figure E3

E4 — Open My Bookings offline, never loaded before

Expected result	Empty state shown, no crash
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

A screenshot of the message or screen shown

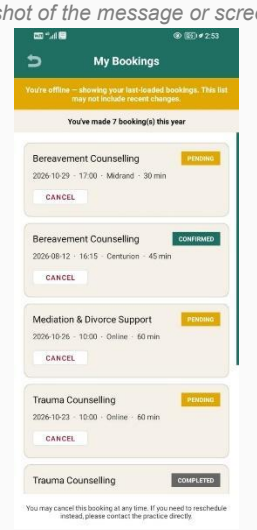


Figure E4

E5 — Holiday API unreachable

Expected result	Booking still works on open-day and hours rules; toast notes holidays couldn't be checked
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

A screenshot of the message or screen shown

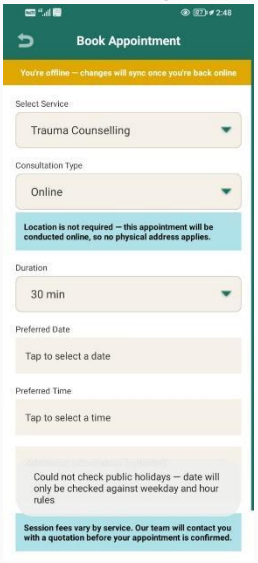


Figure E5

E6 — Search with no matching results

Expected result	Empty state shown, no crash
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

A screenshot of the message or screen shown

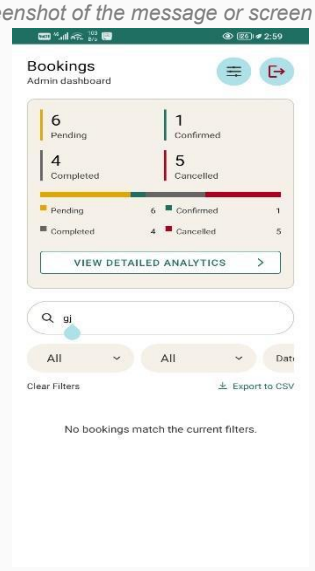


Figure E6

E7 — Export CSV with zero bookings in the filtered list

Expected result	Export still runs with a header row only
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

A screenshot of the message or screen shown

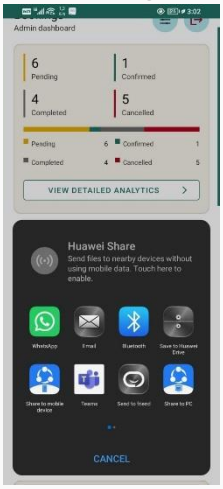


Figure E7

E8 — Back button on a non-root screen (e.g. Book Appointment)

Expected result	Normal back navigation — the 'press again to exit' behaviour applies only to Home/Admin Dashboard
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

A screenshot of the message or screen shown

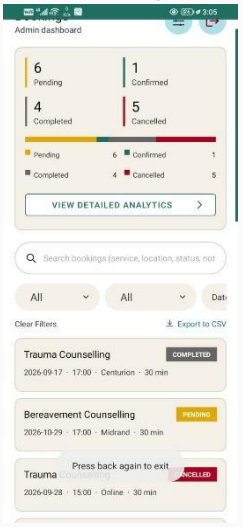


Figure E8

6. Team test plans

Test plans assigned to each group member. Every test is PENDING until it is run; mark PASS or FAIL and insert the evidence under it. Each member's evidence pack is described at the start of their section.

6.1 UX/UI testing — Faith Murau

Goal

Prove the app looks consistent, is easy to understand, and gives clear feedback. Test on at least two screen sizes and in both light and dark mode (Settings has a dark mode switch).

Evidence pack

Side-by-side screenshots of every screen in light and dark mode, a table of every message with the clear/vague/technical rating, and a short list of recommended UI improvements.

Visual consistency

UX-01

Owner	Faith Murau
Steps / what to check	Colour palette across all 22 screens
Expected result	Same primary colour, background and text colours everywhere; status colours consistent (pending, confirmed, cancelled, completed each has one colour).
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

Light-mode and dark-mode screenshots (and a second screen size where relevant)

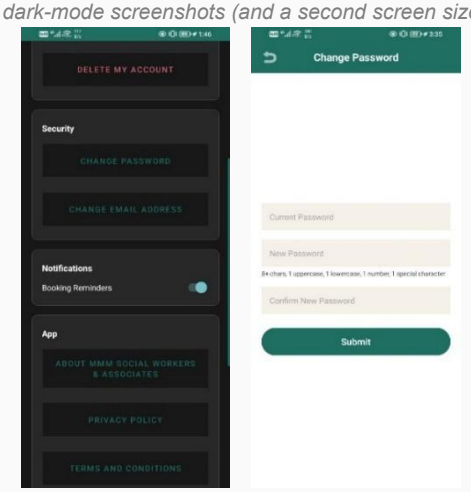


Figure UX-01

UX-02

Owner	Faith Murau
Steps / what to check	Fonts and sizes
Expected result	Titles, body text and button labels use consistent sizes; no text smaller than readable (about 12sp).

Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

Light-mode and dark-mode screenshots (and a second screen size where relevant)

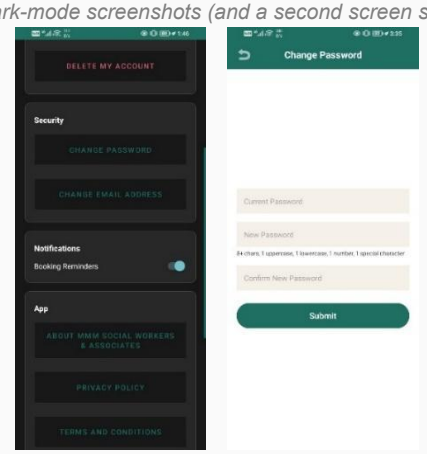


Figure UX-02

UX-03

Owner	Faith Murau
Steps / what to check	Buttons
Expected result	Primary buttons look the same on every screen; disabled buttons look disabled (for example Submit while loading).
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

Light-mode and dark-mode screenshots (and a second screen size where relevant)

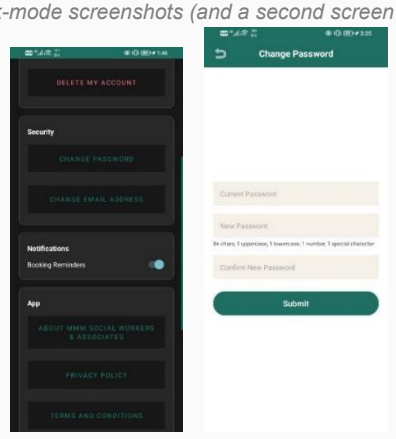


Figure UX-03

UX-04

Owner	Faith Murau
Steps / what to check	Spacing and alignment

Expected result	Equal margins left and right, no overlapping or clipped text, nothing cut off at the bottom.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

Light-mode and dark-mode screenshots (and a second screen size where relevant)

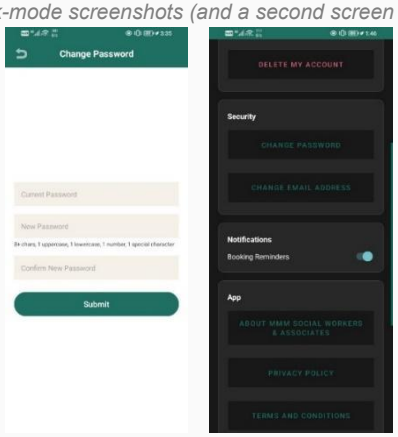


Figure UX-04

UX-05

Owner	Faith Murau
Steps / what to check	Dark mode
Expected result	Switch on in Settings. Every screen stays readable: no dark text on dark background, no white blocks. Setting survives closing and reopening the app.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

Light-mode and dark-mode screenshots (and a second screen size where relevant)

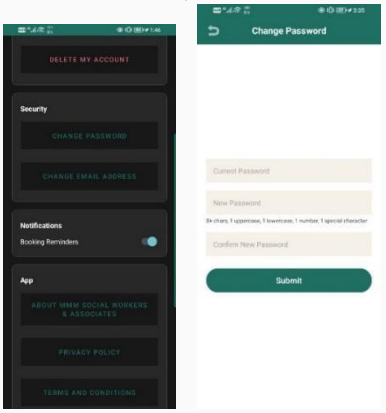


Figure UX-05

UX-06

Owner	Faith Murau

Steps / what to check	Screen sizes and rotation
Expected result	Small phone and tablet both look correct. Rotate on Register, Book Appointment and Admin Dashboard: fields stay filled and nothing crashes.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

Light-mode and dark-mode screenshots (and a second screen size where relevant)

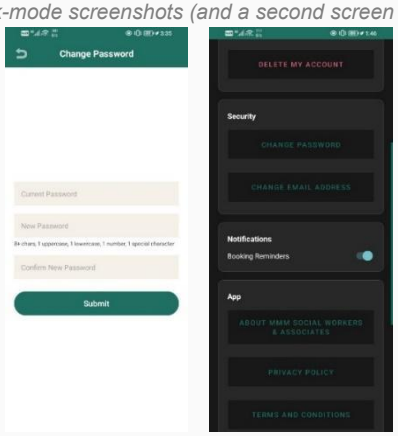


Figure UX-06

UX-07

Owner	Faith Murau
Steps / what to check	Touch targets
Expected result	Buttons and icons are easy to tap with a thumb (about 48dp).
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

Light-mode and dark-mode screenshots (and a second screen size where relevant)

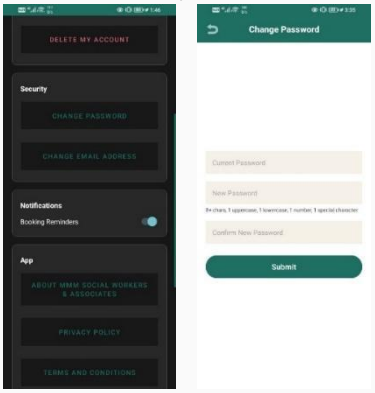


Figure UX-07

UX-08

Owner	Faith Murau

Steps / what to check	Branding
Expected result	App name, logo and colours match on Welcome, Login, Home and About.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

Light-mode and dark-mode screenshots

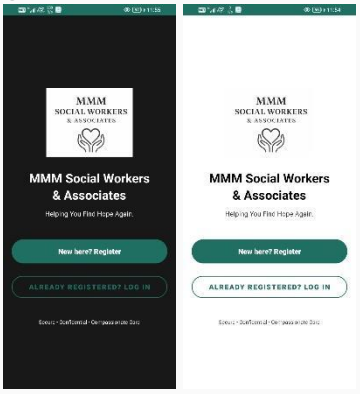


Figure UX-08

Clarity and ease of use

UX-09

Owner	Faith Murau
Steps / what to check	First-time user test: ask someone who has never seen the app to register, book and cancel without help. Time them and note where they hesitate.
Expected result	The task is completed without help; hesitation points are recorded.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

Light-mode and dark-mode screenshots

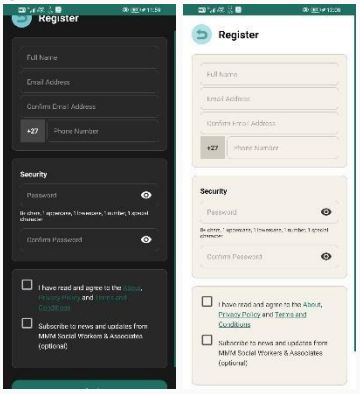


Figure UX-09

UX-10

Owner	Faith Murau

Steps / what to check	Labels and hints
Expected result	Every input has a clear label or hint (phone shows that the +27 prefix is fixed, password requirements are visible).
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

Light-mode and dark-mode screenshots

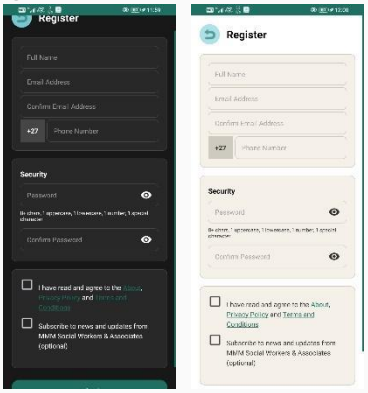


Figure UX-10

UX-11

Owner	Faith Murau
Steps / what to check	Booking screen flow
Expected result	Service, type, location (only when In-Person), duration, date, time, notes: order makes sense, Online shows a note instead of a location.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

Light-mode and dark-mode screenshots

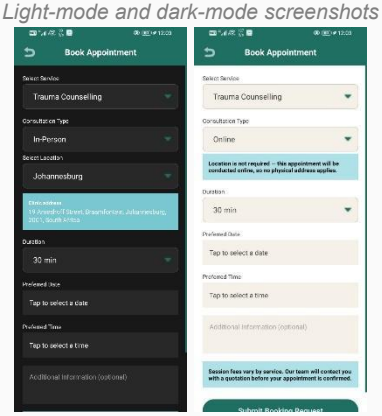


Figure UX-11

UX-12

Owner	Faith Murau

Steps / what to check	Empty states: a new client opens My Bookings; the Admin Dashboard with no bookings
Expected result	A friendly empty message appears on each.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

Light-mode and dark-mode screenshots

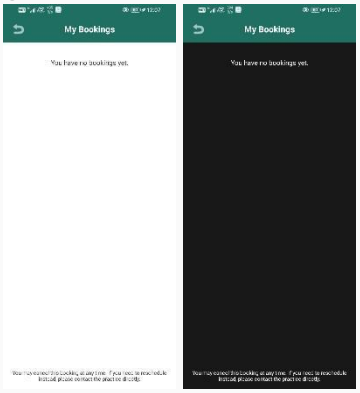


Figure UX-12

UX-13

Owner	Faith Murau
Steps / what to check	Loading feedback
Expected result	Progress indicator shows while booking, logging in and loading lists; the user is never left looking at a frozen screen.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

Light-mode and dark-mode screenshots

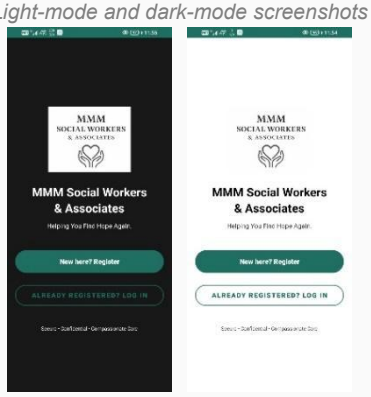


Figure UX-13

UX-14

Owner	Faith Murau

Steps / what to check	Offline banner: turn off data, then turn it back on
Expected result	A banner appears on supported screens; booking is blocked with a clear message. When data is back on, the banner disappears.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

Light-mode and dark-mode screenshots

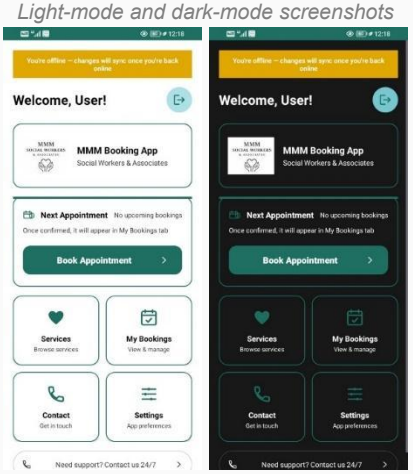


Figure UX-14

Error and confirmation messages (record the exact message and rate it: clear, vague or technical)

UX-15

Owner	Faith Murau
Steps / what to check	Register with a one-word name
Expected result	Inline error that two names are needed.
Message shown / rating	Exact message: **“Enter your full name(at least 2 words, 2+ letters each)”**
Result	**PASS and CLEAR**
Device / date	Phone and Table Date: 15/10/2026

Screenshot of the message, plus the exact wording

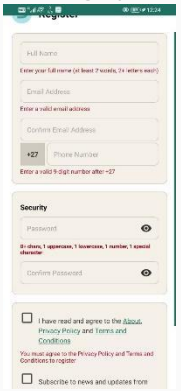


Figure UX-15

UX-16

Owner	Faith Murau
Steps / what to check	Register with 8 digits in phone
Expected result	Inline phone error (9 digits required after +27).
Message shown / rating	Exact message: **“Enter a valid 9-digit number after +27”**
Result	**PASS and CLEAR**
Device / date	Phone and Table Date: 15/10/2026

Screenshot of the message, plus the exact wording



Figure UX-16

UX-17

Owner	Faith Murau
Steps / what to check	Register with mismatched email or password confirmation
Expected result	Inline mismatch errors.
Message shown / rating	Exact message: **“Enter a valid email address”**
Result	**PASS and CLEAR**
Device / date	Phone and Table Date: 15/10/2026

Screenshot of the message, plus the exact wording



Figure UX-17

UX-18

Owner	Faith Murau
Steps / what to check	Register without ticking consent
Expected result	Red consent error, no account created.
Message shown / rating	Exact message: **“You must agree to the Privacy Policy and Terms and Conditions to Register”**
Result	**PASS and CLEAR**
Device / date	Phone and Table Date: 15/10/2026

Screenshot of the message, plus the exact wording

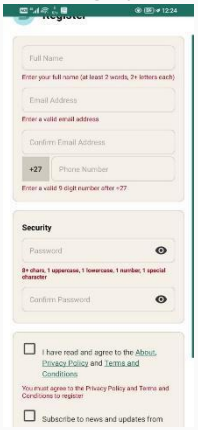


Figure UX-18

UX-19

Owner	Faith Murau
Steps / what to check	Register with a weak password
Expected result	The password rules line turns red and bold next to the field (no toast), consistent with the other fields.
Message shown / rating	Exact message: **“8+chars, 1 uppercase, 1 lowercase, 1 number, 1 special character”**
Result	**PASS and CLEAR**
Device / date	Phone and Table Date: 15/10/2026

Screenshot of the message, plus the exact wording

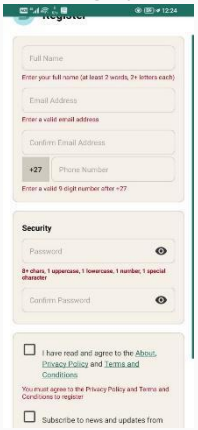


Figure UX-19

UX-20

Owner	Faith Murau
Steps / what to check	Login with empty fields
Expected result	'This field is required' appears on each empty field.
Message shown / rating	Exact message: **“Fill in”**
Result	**PASS and CLEAR**

Device / date	Phone and Table Date: 15/10/2026

Screenshot of the message, plus the exact wording

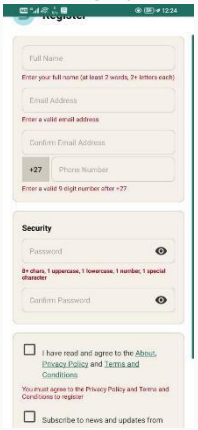


Figure UX-20

UX-21

Owner	Faith Murau
Steps / what to check	Login with wrong password
Expected result	Clear failure message that does not reveal which part was wrong.
Message shown / rating	Exact message: **“Login failed”**
Result	**PASS and CLEAR**
Device / date	Phone and Table Date: 15/10/2026

Screenshot of the message, plus the exact wording

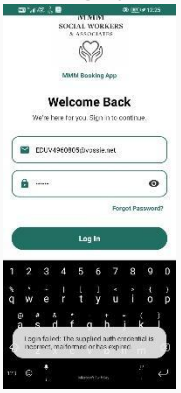


Figure UX-21

UX-22

Owner	Faith Murau
Steps / what to check	Booking on a closed day, a public holiday, and outside hours
Expected result	Specific messages naming the open days, the holiday, or the opening and closing times.

Message shown / rating	Exact message: **“Booking on a closed day, a public holiday, and outside hours”**
Result	**PASS and CLEAR**
Device / date	Phone and Table Date: 15/10/2026

Screenshot of the message, plus the exact wording



Figure UX-22

UX-23

Owner	Faith Murau
Steps / what to check	Booking on a taken time
Expected result	Clear message that the time is already reserved.
Message shown / rating	Exact message: **“Time is already reserved.”**
Result	**PASS and CLEAR**
Device / date	Phone and Table Date: 15/10/2026

Screenshot of the message, plus the exact wording



Figure UX-23

UX-24

Owner	Faith Murau
Steps / what to check	Successful booking, cancel, decline, delete, paid toggle

Expected result	A confirmation message appears for each, and destructive actions (cancel, decline, delete) show a confirmation dialog first.
Result	**PASS and CLEAR**
Device / date	Phone and Table Date: 15/10/2026

Screenshot of the message, plus the exact wording

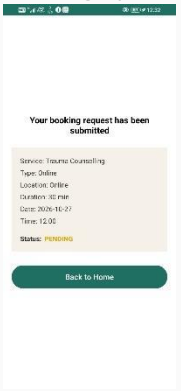


Figure UX-24

6.2 Screen and navigation testing — Khauhelo Moqechane

Goal
Every screen opens, every button does something, and the back stack behaves.

Evidence pack
The completed inventory table, a navigation flow diagram (draw the screen graph that you actually observed), a list of broken or missing items, and a short screen recording of the full client journey and the admin journey.

Screen inventory (open each screen once and tick it off)

Screen	Reached from	Notes	Opened
Splash / router	App launch	Routes to Welcome, client Home or admin Dashboard depending on session	☐
Welcome	Splash (logged out)	Opens with working Login and Register buttons	☐
Login	Welcome	Opens. Login works for both client and admin.	☐
Register	Welcome	Opens. Form and validation errors show.	☐
Forgot Password	Login	Opens. Reset email request works.	☐
Home	Login (client)	Opens after client login.	☐
Services	Home	Opens and lists the services.	☐
Book Appointment	Home or Services	Opens. Booking form works.	☐
Booking Confirmation	After a successful booking	Opens after submitting a valid booking.	☐
My Bookings	Home	Opens and lists the client's bookings.	☐
Contact	Home	Opens.	☐
Settings	Home or menu	Opens. Dark mode and reminders available.	☐
Change Password / Change Email	Settings	Both open from Settings.	☐
FAQ / About / Privacy / Terms	Settings or Register (terms)	All four open.	☐
Admin Dashboard	Login (admin)	Opens after admin login, not the client Home.	☐
Booking Detail	Dashboard list item	Opens when a booking is tapped.	☐
Practice Information	Dashboard	Opens.	☐
Admin Analytics	Dashboard	Opens.	☐

Navigation tests

NAV-01

Owner	Khauhelo Moqechane
Steps / what to check	Fresh install, open app
Expected result	Welcome screen with working Login and Register buttons.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

Screenshots



Figure NAV-01

NAV-02

Owner	Khauhelo Moqechane
Steps / what to check	Login with a verified client
Expected result	Lands on Home. Pressing back does not return to Login.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

Screenshots or a short screen recording of the steps and the result

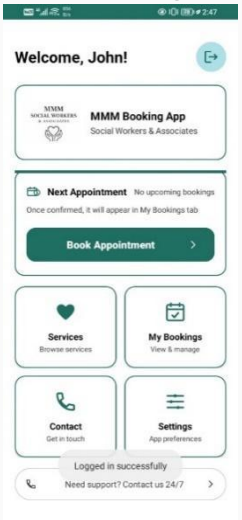


Figure NAV-02

NAV-03

Owner	Khauhelo Moqechane
Steps / what to check	Login with admin
Expected result	Lands on Admin Dashboard, not client Home.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

Screenshots or a short screen recording of the steps and the result

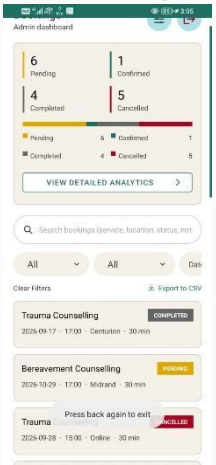


Figure NAV-03

NAV-04

Owner	Khauhelo Moqechane
Steps / what to check	Close app (swipe away) while logged in, reopen
Expected result	Goes straight to the correct home screen for the role, no login required.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

Screenshots

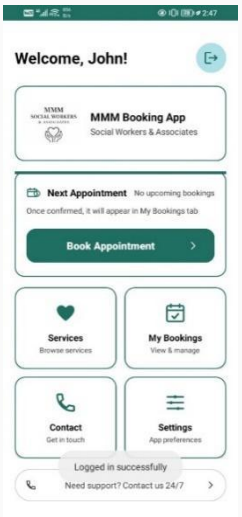


Figure NAV-04

NAV-05

Owner	Khauhelo Moqechane
Steps / what to check	Log out, press back
Expected result	Cannot return to a logged-in screen.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

Screenshots



Figure NAV-05

NAV-06

Owner	Khauhelo Moqechane
Steps / what to check	Welcome to Register to Login using on-screen links, then back
Expected result	Back moves one step at a time with no crash or blank screen.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

Screenshots



Figure NAV-06

NAV-07

Owner	Khauhelo Moqechane
Steps / what to check	Register successfully

Expected result	Verification message shown, user returned to a sensible screen (Login), not left in a half-logged-in state.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

Screenshot

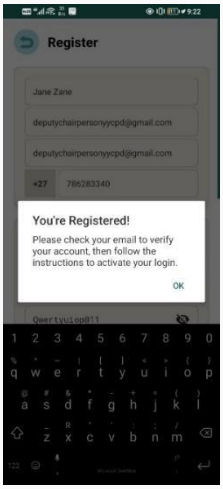


Figure NAV-07

NAV-08

Owner	Khauhelo Moqechane
Steps / what to check	Login, then Forgot Password, then back
Expected result	Returns to Login with fields intact.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

Screenshot



Figure NAV-08

NAV-09

Owner	Khauhelo Moqechane
Steps / what to check	Home, then Book Appointment, then submit

Expected result	Booking Confirmation opens; back from Confirmation goes Home or My Bookings, not back into the filled booking form (the form is closed on submit).
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

Screenshot

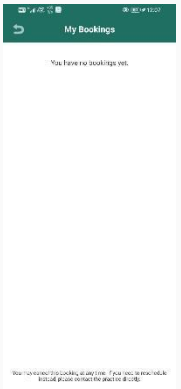


Figure NAV-09

NAV-10

Owner	Khauhelo Moqechane
Steps / what to check	Home, then My Bookings, cancel a booking, then back
Expected result	List already updated (live listener), back goes Home.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

Screenshot

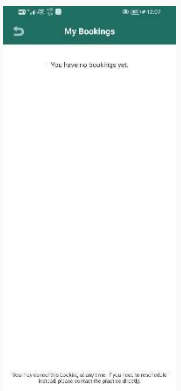


Figure NAV-10

NAV-11

Owner	Khauhelo Moqechane
Steps / what to check	Settings, then Change Password, back; Change Email, back
Expected result	Returns to Settings each time.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

Screenshot



Figure NAV-11

NAV-12

Owner	Khauhelo Moqechane
Steps / what to check	Terms/Privacy opened from Register, then back
Expected result	Registration form keeps its typed values.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

Screenshot

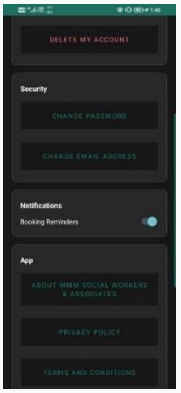


Figure NAV-12

NAV-13

Owner	Khauhelo Moqechane
Steps / what to check	Admin Dashboard, then Booking Detail, confirm, back
Expected result	Dashboard shows the updated status.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

Screenshot

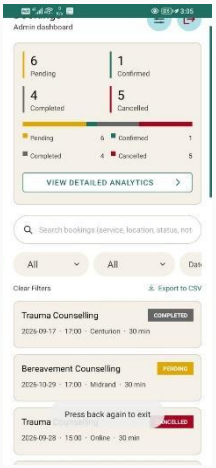


Figure NAV-13

NAV-14

Owner	Khauhelo Moqechane
Steps / what to check	Admin Dashboard: press back once, then again quickly
Expected result	First press shows an exit hint, second exits (double-press behaviour).
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

Screenshot

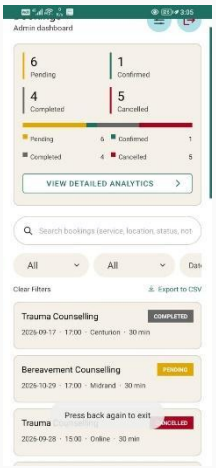


Figure NAV-14

NAV-15

Owner	Khauhelo Moqechane
Steps / what to check	Rotate device on every major screen
Expected result	No crash; data preserved or sensibly reloaded.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

Screenshot

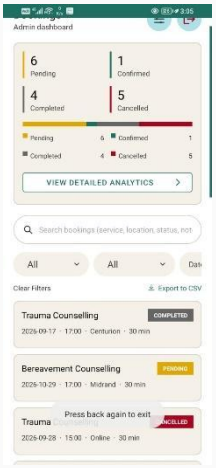


Figure NAV-15

NAV-16

Owner	Khauhelo Moqechane
Steps / what to check	Rapidly tap a button twice (Login, Submit Booking)
Expected result	Only one action happens (button disabled while loading); no duplicate booking.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

Screenshot

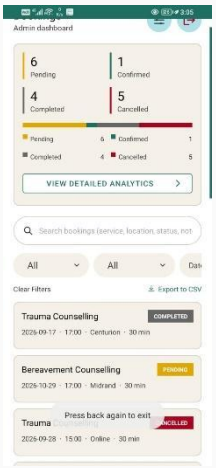


Figure NAV-16

NAV-17

Owner	Khauhelo Moqechane
Steps / what to check	Open app from a notification (reminder or booking submitted)
Expected result	Opens the app without a crash.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

Screenshot

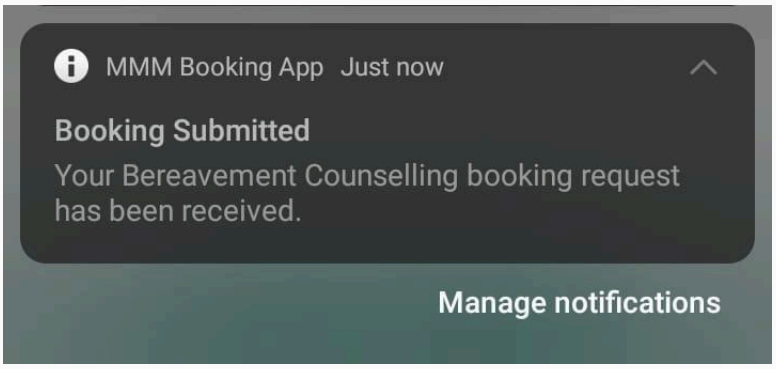


Figure NAV-17

NAV-18

Owner	Khauhelo Moqechane
Steps / what to check	Turn off data, then tap navigation items
Expected result	Screens open; the offline banner shows; no crash.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

Screenshot

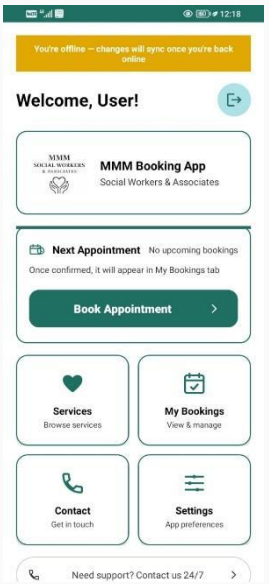


Figure NAV-18

6.3 Database, authentication and booking logic — Ignatius Mayibongwe Khumalo

Goal

Prove the backend rules work, data is stored correctly and access control cannot be bypassed. Keep the Firebase Console open beside the phone and check the actual documents after every action.

Evidence pack

Firestore Console screenshots before and after each action, the Rules Playground result for every PM test, and a table of the stored fields against the data design from Deliverable 1.

Validation rules to test against

Password: at least 8 characters with upper case, lower case, a digit and a special character. Phone: +27 fixed prefix and exactly 9 digits. Name: at least two words of at least two letters.

Authentication (DB-01 to DB-13)

DB-01

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Register Client A with valid details
Expected result	Firebase Authentication shows the new user; users/{uid} exists with name, email, phone and role: "client"; verification email arrives.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

App screenshot plus the matching Firebase Console view (Authentication / Firestore)

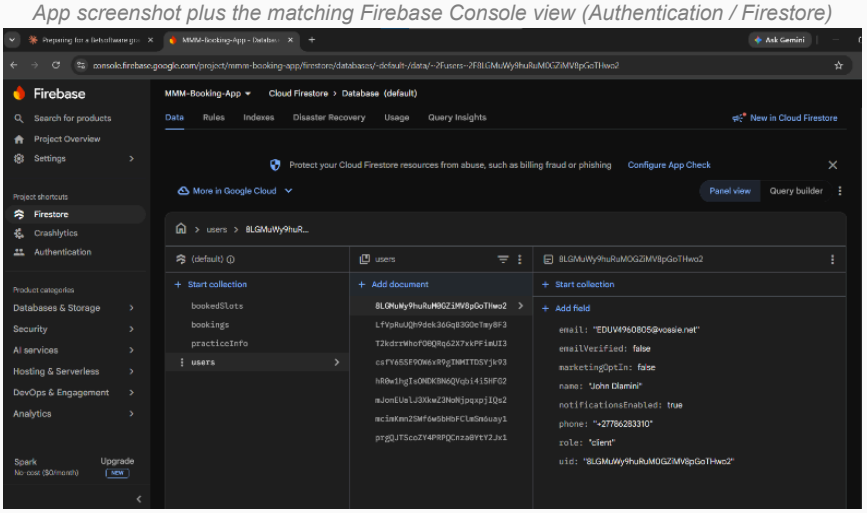


Figure DB-01

DB-02

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Login before verifying email
Expected result	Blocked, with an option to resend the verification email.
Result	**PASS**

Device / date	Phone and Table Date: 15/10/2026

App screenshot plus the matching Firebase Console view (Authentication / Firestore)

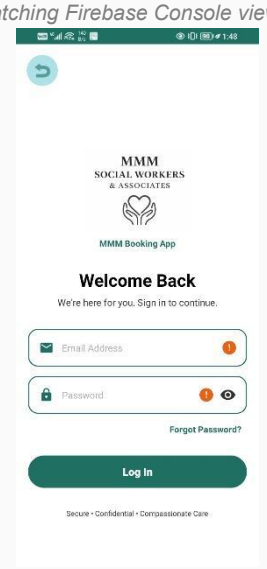


Figure DB-02

DB-03

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Verify via email link, log in
Expected result	Succeeds and routes to Home.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

App screenshot plus the matching Firebase Console view (Authentication / Firestore)

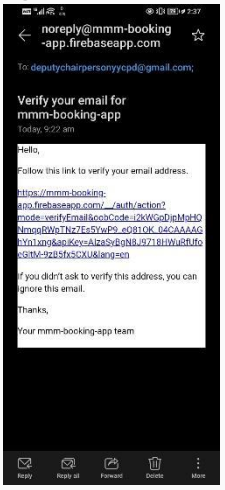


Figure DB-03

DB-04

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Wrong password; unregistered email

Expected result	Login fails with a message; no session created.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

App screenshot plus the matching Firebase Console view (Authentication / Firestore)



Figure DB-04

DB-05

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Register with an already used email
Expected result	Clear failure, no duplicate user document.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

App screenshot plus the matching Firebase Console view (Authentication / Firestore)



Figure DB-05

DB-06

Owner	Ignatius Mayibongwe Khumalo

Steps / what to check	Boundary data: password with 7 chars, no uppercase, no digit, no special; phone 8 and 10 digits; name "A B"
Expected result	Each is rejected.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

App screenshot plus the matching Firebase Console view (Authentication / Firestore)

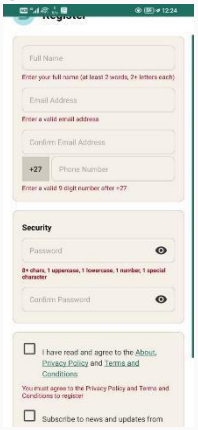


Figure DB-06

DB-07

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Email without a proper domain, for example abc@ or a@b
Expected result	Rejected with the email error (the email format is checked, not just the @ sign).
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

App screenshot plus the matching Firebase Console view (Authentication / Firestore)

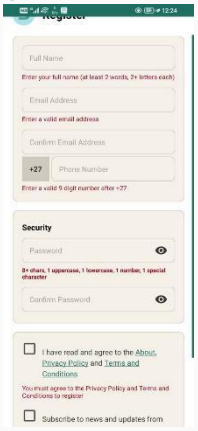


Figure DB-07

DB-08

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Forgot Password with a registered email

Expected result	Reset email received and the new password works.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

App screenshot plus the matching Firebase Console view (Authentication / Firestore)

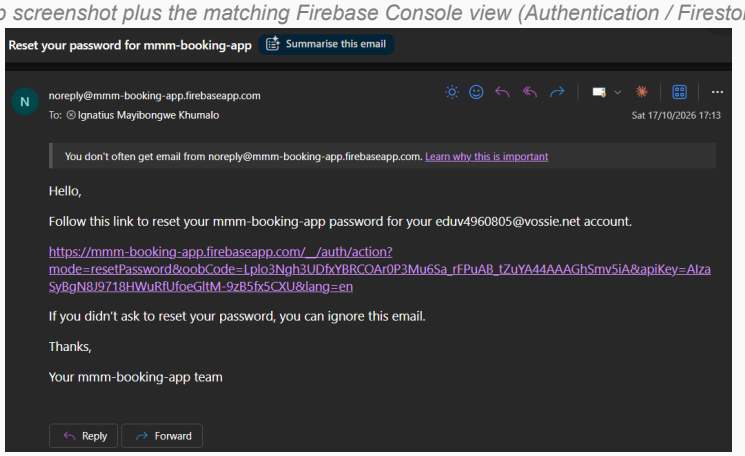


Figure DB-08

DB-09

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Change Password with correct old password; with wrong old password
Expected result	Succeeds / rejected. Login with the new password works.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

App screenshot plus the matching Firebase Console view (Authentication / Firestore)



Figure DB-09

DB-10

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Change Email
Expected result	Verification flow works and the users document email stays consistent.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

App screenshot plus the matching Firebase Console view (Authentication / Firestore)

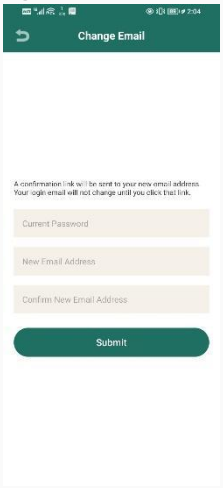


Figure DB-10

DB-11

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Logout, then relaunch
Expected result	Back at Welcome, no session.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

App screenshot plus the matching Firebase Console view (Authentication / Firestore)



Figure DB-11

DB-12

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Register while offline
Expected result	Fails with a message and no orphaned account is left in Firebase Authentication.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

App screenshot plus the matching Firebase Console view (Authentication / Firestore)

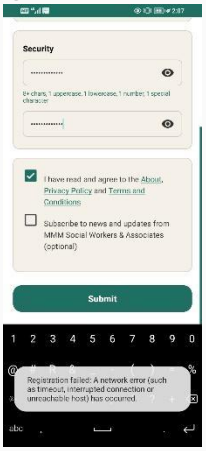


Figure DB-12

DB-13

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Login as the admin account
Expected result	Routed to the admin dashboard; client accounts never reach it.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

App screenshot plus the matching Firebase Console view (Authentication / Firestore)

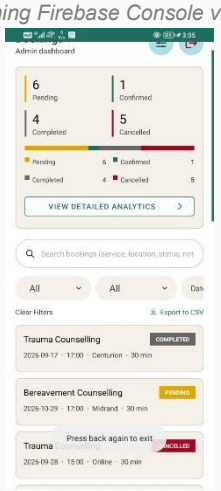


Figure DB-13

Data saved and retrieved (DB-14 to DB-21)

DB-14

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Client A books In-Person, 60 min, with notes
Expected result	bookings/{id} has clientId = A's uid, service, consultationType in-person, location, duration 60, date yyyy-MM-dd, time HH:mm, additionalInfo, status pending, paid false, createdAt.
Result	**PASS**

Device / date	Phone and Table Date: 15/10/2026

App screenshot plus the Firestore document showing the stored fields

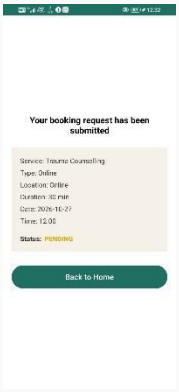


Figure DB-14

DB-15

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Same booking in My Bookings
Expected result	All details match what was entered.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

App screenshot plus the Firestore document showing the stored fields

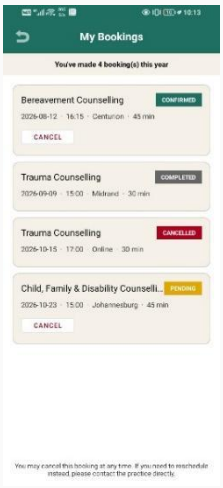


Figure DB-15

DB-16

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Book Online
Expected result	location is null and consultationType is online.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

INSERT SCREENSHOT FOR DB-16 HERE

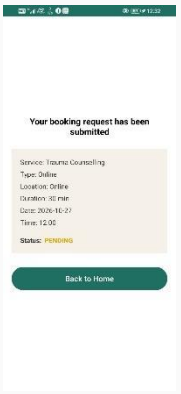


Figure DB-16

DB-17

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Admin opens Booking Detail
Expected result	Shows client name, phone and email read from users, plus all booking fields.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

App screenshot plus the Firestore document showing the stored fields

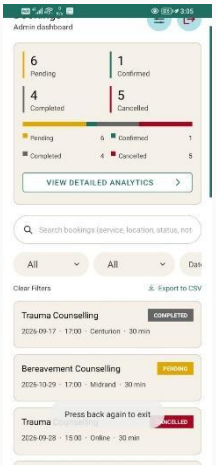


Figure DB-17

DB-18

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Admin confirms, marks paid, marks completed
Expected result	Firestore status and paid change each time and Client A sees the new status live without refreshing.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

App screenshot plus the Firestore document showing the stored fields

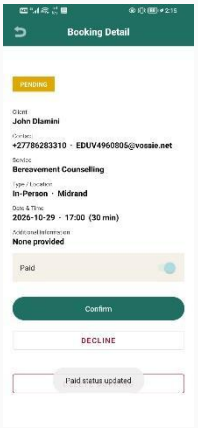


Figure DB-18

DB-19

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Edit Practice Information (days, hours)
Expected result	practiceInfo document updates and the Book Appointment screen immediately uses the new hours.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

App screenshot plus the Firestore document showing the stored fields



Figure DB-19

DB-20

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Practice hours validation: end before start; malformed times; no open days
Expected result	Each rejected with a message; nothing saved.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

App screenshot plus the Firestore document showing the stored fields



Figure DB-20

DB-21

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Settings: toggle reminders and dark mode
Expected result	Preference saved on the account/device and still applied after restart.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

App screenshot plus the Firestore document showing the stored fields

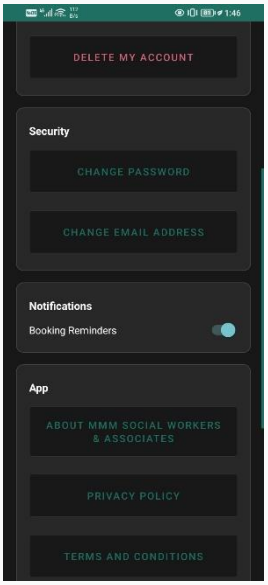


Figure DB-21

6.4 Booking logic — Ignatius Mayibongwe Khumalo

How slots work
A booking reserves one bookedSlots document per 15-minute block, named date_time_location (location is "online" for online bookings). A 60-minute booking at 10:00 reserves 10:00, 10:15, 10:30 and 10:45.

BK-01

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Book 10:00 for 60 min

Expected result	Four slot documents appear with Client A's uid as clientId.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

App message plus the Firestore bookings / bookedSlots result (use two phones where needed)

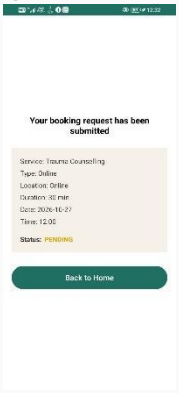


Figure BK-01

BK-02

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Client B books the same location, same date, 10:00
Expected result	Rejected: time already reserved. No booking or slots created for B.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

App message plus the Firestore bookings / bookedSlots result (use two phones where needed)

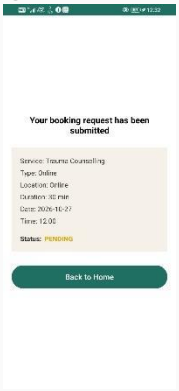


Figure BK-02

BK-03

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Client B books 10:30, 30 min (overlaps the middle of A's)
Expected result	Rejected. This is the key overlap test; the original version of the app wrongly allowed it.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

App message plus the Firestore bookings / bookedSlots result (use two phones where needed)

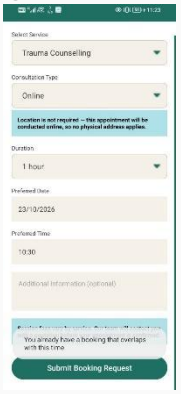


Figure BK-03

BK-04

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Client B books 09:00 for 60 min (ends exactly at 10:00)
Expected result	Accepted: back-to-back appointments are allowed.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

App message plus the Firestore bookings / bookedSlots result (use two phones where needed)

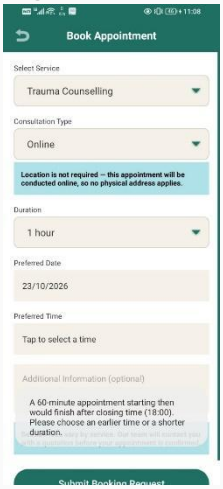


Figure BK-04

BK-05

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Client B books 11:00
Expected result	Accepted (A's last block is 10:45 to 11:00).
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

App message plus the Firestore bookings / bookedSlots result (use two phones where needed)

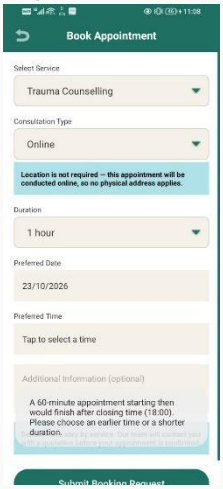


Figure BK-05

BK-06

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Client B books the same time at a different location
Expected result	Accepted: each location is its own calendar.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

App message plus the Firestore bookings / bookedSlots result (use two phones where needed)

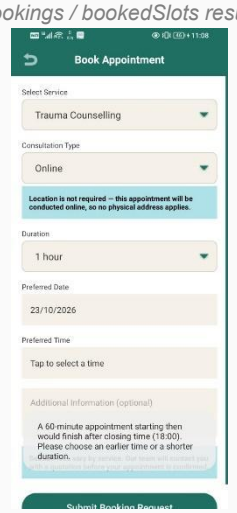


Figure BK-06

BK-07

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Client A books another appointment overlapping her own
Expected result	Rejected with the 'overlaps your own booking' message.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

App message plus the Firestore bookings / bookedSlots result (use two phones where needed)

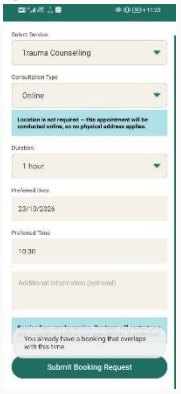


Figure BK-07

BK-08

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Two phones submit the same slot at the same moment (repeat three times)
Expected result	Only one succeeds (transaction).
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

App message plus the Firestore bookings / bookedSlots result (use two phones where needed)

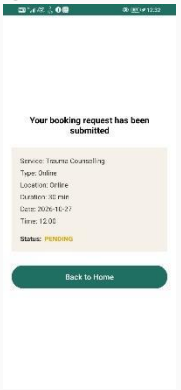


Figure BK-08

BK-09

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Book on a closed day (for example Sunday)
Expected result	Rejected, message names the open days.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

App message plus the Firestore bookings / bookedSlots result (use two phones where needed)



Figure BK-09

BK-10

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Book on a South African public holiday (for example 25 December, 1 January), and a holiday in next year
Expected result	Rejected with the holiday message; the next-year date confirms the per-year fetch.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

App message plus the Firestore bookings / bookedSlots result (use two phones where needed)



Figure BK-10

BK-11

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Book before opening time; at 16:45 for 60 min when closing is 17:00
Expected result	Rejected with the correct reason (before opening; runs past closing).
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

App message plus the Firestore bookings / bookedSlots result (use two phones where needed)

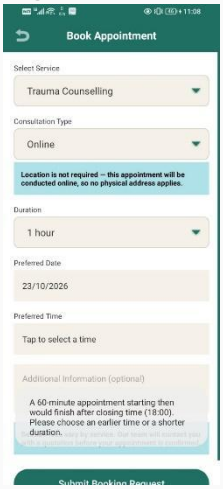


Figure BK-11

BK-12

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Choose 16:00 with 30 min, then change duration to 60 min while closing is 17:00
Expected result	If it no longer fits, the time is cleared and the user told.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

App message plus the Firestore bookings / bookedSlots result (use two phones where needed)

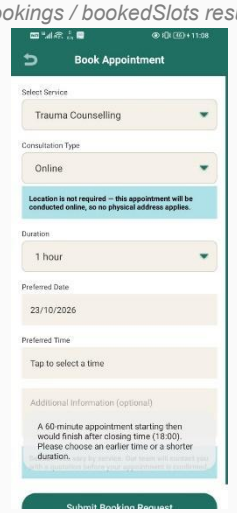


Figure BK-12

BK-13

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Choose a time with minutes not on the quarter hour (10:07)
Expected result	Rejected.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

App message plus the Firestore bookings / bookedSlots result (use two phones where needed)

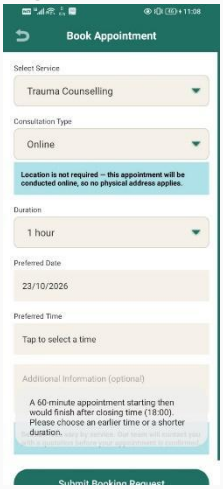


Figure BK-13

BK-14

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Book with no date selected; with no time
Expected result	Rejected with a message.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

App message plus the Firestore bookings / bookedSlots result (use two phones where needed)

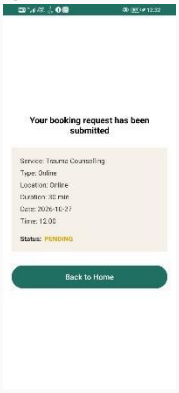


Figure BK-14

BK-15

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Cancel Client A's booking (confirm dialog)
Expected result	Booking status becomes cancelled; all four slot documents are deleted in the same operation; reminder cancelled.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

App message plus the Firestore bookings / bookedSlots result (use two phones where needed)

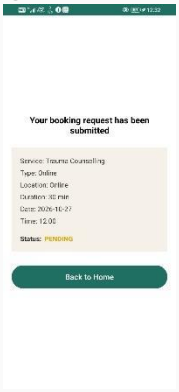


Figure BK-15

BK-16

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Client B immediately books the freed time
Expected result	Accepted.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

App message plus the Firestore bookings / bookedSlots result (use two phones where needed)

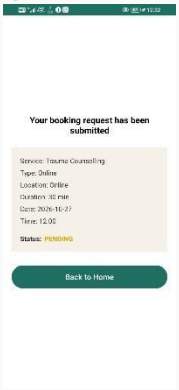


Figure BK-16

BK-17

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Cancel an already cancelled or completed booking
Expected result	Cancel button not offered, or a clear 'cannot be cancelled' message.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

App message plus the Firestore bookings / bookedSlots result (use two phones where needed)



Figure BK-17

BK-18

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Admin declines a pending booking (confirm dialog)
Expected result	Same result as BK-15 and the record is kept as cancelled.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

App message plus the Firestore bookings / bookedSlots result (use two phones where needed)

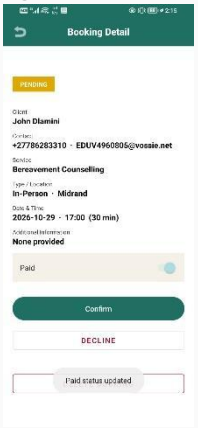


Figure BK-18

BK-19

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Admin deletes a booking (confirm dialog)
Expected result	Booking document is gone; slots are freed if it was still active; Client A's list updates.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

App message plus the Firestore bookings / bookedSlots result (use two phones where needed)

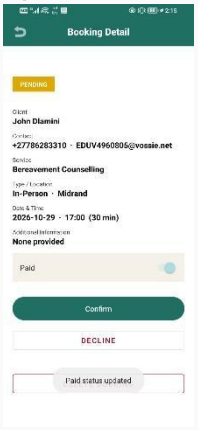


Figure BK-19

BK-20

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Delete an old cancelled booking whose time has since been re-booked by someone else
Expected result	The new booking's slots are not removed.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

App message plus the Firestore bookings / bookedSlots result (use two phones where needed)

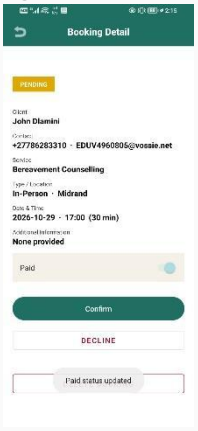


Figure BK-20

BK-21

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Go offline, then submit a booking and cancel a booking
Expected result	Both blocked with the offline message; no partial data in Firestore.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

App message plus the Firestore bookings / bookedSlots result (use two phones where needed)

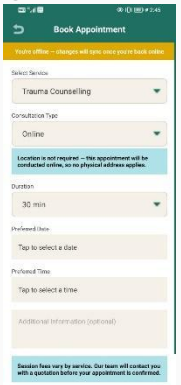


Figure BK-21

BK-22

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Book an Online appointment at 10:00, then another client books Online at 10:00
Expected result	Rejected, because all online bookings share one calendar. Record this as a design limitation (one online appointment at a time).
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

App message plus the Firestore bookings / bookedSlots result (use two phones where needed)

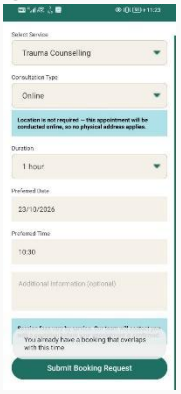


Figure BK-22

6.5 Database permissions and access control — Ignatius Mayibongwe Khumalo

How to run
Run these two ways: (a) by using the app as the relevant user, and (b) in the Firebase Rules Playground, where you can simulate an authenticated request for a chosen uid. The Playground result is the proof for the report.

PM-01

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Unauthenticated read of bookings or users
Expected result	Denied.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

Rules Playground result screenshot (simulated request and Allowed/Denied outcome)

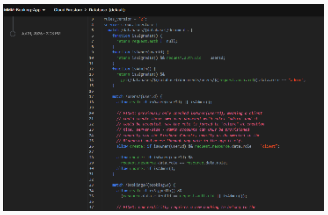


Figure PM-01

PM-02

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Unauthenticated read of practiceInfo
Expected result	Allowed (public by design, needed for the app before login).
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

Rules Playground result screenshot (simulated request and Allowed/Denied outcome)

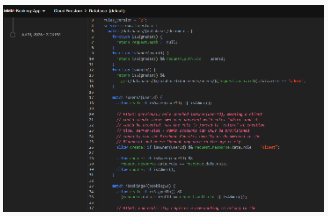


Figure PM-02

PM-03

Owner	Ignatius Mayibongwe Khumalo

Steps / what to check	Client A reads users/{A}
Expected result	Allowed.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

Rules Playground result screenshot (simulated request and Allowed/Denied outcome)

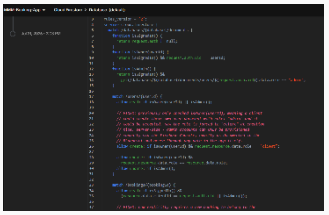


Figure PM-03

PM-04

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Client A reads users/{B}
Expected result	Denied.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

Rules Playground result screenshot (simulated request and Allowed/Denied outcome)

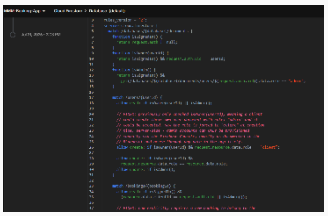


Figure PM-04

PM-05

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Admin reads any users document
Expected result	Allowed.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

Rules Playground result screenshot (simulated request and Allowed/Denied outcome)

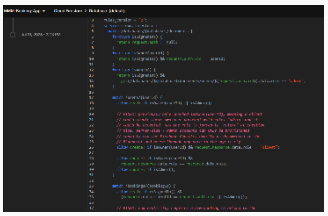


Figure PM-05

PM-06

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Create users/{A} with role: "admin"
Expected result	Denied (create requires role client). This was a privilege-escalation hole in the first version.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

Rules Playground result screenshot (simulated request and Allowed/Denied outcome)

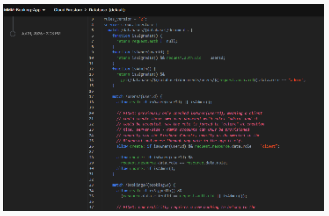


Figure PM-06

PM-07

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Client A updates own profile and changes role to admin
Expected result	Denied.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

Rules Playground result screenshot (simulated request and Allowed/Denied outcome)

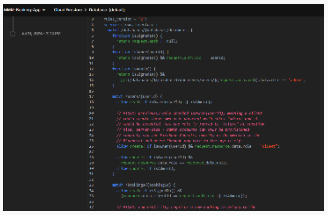


Figure PM-07

PM-08

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Client A reads own booking
Expected result	Allowed.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

Rules Playground result screenshot (simulated request and Allowed/Denied outcome)

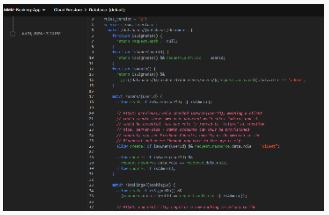


Figure PM-08

PM-09

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Client A reads Client B's booking
Expected result	Denied.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

Rules Playground result screenshot (simulated request and Allowed/Denied outcome)

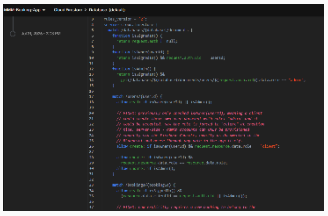


Figure PM-09

PM-10

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Client A creates a booking with status confirmed, or paid true, or another user's clientId
Expected result	Denied.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

Rules Playground result screenshot (simulated request and Allowed/Denied outcome)

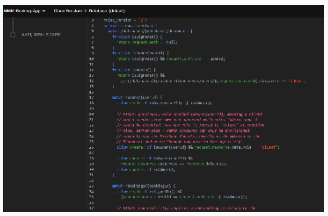


Figure PM-10

PM-11

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Client A updates own booking to status cancelled only
Expected result	Allowed.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

Rules Playground result screenshot (simulated request and Allowed/Denied outcome)

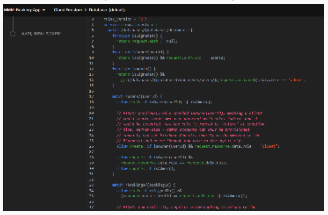


Figure PM-11

PM-12

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Client A updates own booking to confirmed, or sets paid: true, or changes the date
Expected result	Denied.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

Rules Playground result screenshot (simulated request and Allowed/Denied outcome)

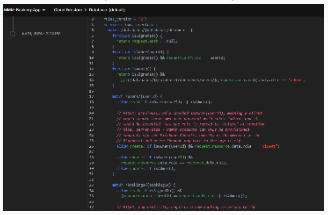


Figure PM-12

PM-13

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Client A deletes a booking
Expected result	Denied.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

Rules Playground result screenshot (simulated request and Allowed/Denied outcome)

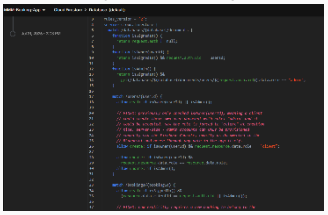


Figure PM-13

PM-14

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Admin updates status and paid; admin deletes a booking
Expected result	Allowed.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

Rules Playground result screenshot (simulated request and Allowed/Denied outcome)

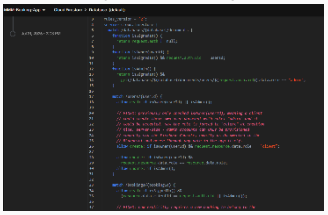


Figure PM-14

PM-15

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Client A creates a slot with clientId = B's uid
Expected result	Denied.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

INSERT SCREENSHOT FOR PM-15 HERE

Rules Playground result screenshot (simulated request and Allowed/Denied outcome)

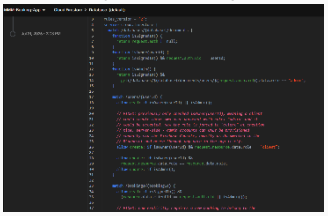


Figure PM-15

PM-16

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Client A deletes Client B's slot
Expected result	Denied.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

Rules Playground result screenshot (simulated request and Allowed/Denied outcome)

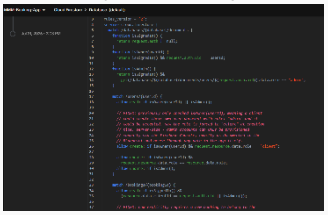


Figure PM-16

PM-17

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Anyone updates (modifies) a slot
Expected result	Denied.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

Rules Playground result screenshot (simulated request and Allowed/Denied outcome)

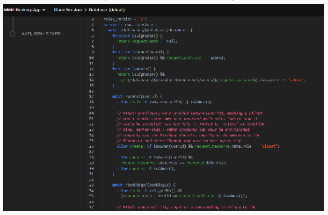


Figure PM-17

PM-18

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Client A writes practiceInfo
Expected result	Denied. Admin write allowed.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

Rules Playground result screenshot (simulated request and Allowed/Denied outcome)

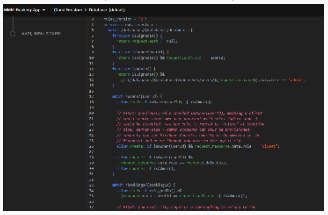


Figure PM-18

6.6 Integration testing — All three members

How to run
Integration tests follow one complete story across the interface, authentication, Firestore and notifications. Run each with two phones (Client on one, Admin on the other) so the live updates can be seen.

INT-01

Owner	All three members
Steps / what to check	New client lifecycle
Expected result	Register, verify email, login, view services, book, see confirmation, see booking in My Bookings, Firestore documents correct.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

Screenshots or a screen recording across both phones, plus the Firestore result



Figure INT-01

INT-02

Owner	All three members
Steps / what to check	Admin approval loop
Expected result	Client books; admin dashboard updates in real time; admin confirms; client's My Bookings changes to confirmed without a refresh; admin marks paid, then completed.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

Screenshots or a screen recording across both phones, plus the Firestore result

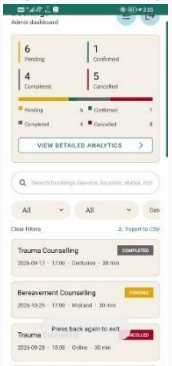


Figure INT-02

INT-03

Owner	All three members
Steps / what to check	Cancellation loop
Expected result	Client cancels; admin dashboard shows cancelled; slots freed; second client can book the same time.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

Screenshots or a screen recording across both phones, plus the Firestore result

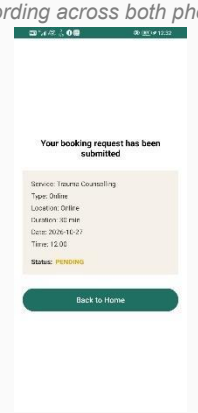


Figure INT-03

INT-04

Owner	All three members
Steps / what to check	Admin changes practice hours
Expected result	Admin changes closing time; client booking screen enforces it on the next attempt.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

Screenshots or a screen recording across both phones, plus the Firestore result



Figure INT-04

INT-05

Owner	All three members

Steps / what to check	Holiday API integration
Expected result	Booking blocked on a public holiday; then block the network and confirm the app warns that holidays could not be loaded and still enforces other rules.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

Screenshots or a screen recording across both phones, plus the Firestore result

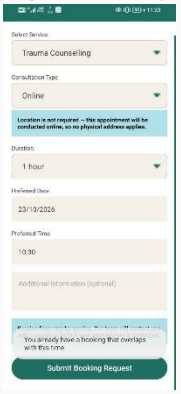


Figure INT-05

INT-06

Owner	All three members
Steps / what to check	Notifications
Expected result	Booking submitted shows an immediate notification; the reminder for the evening before is scheduled; turning reminders off in Settings cancels them; turning on reschedules; cancelling a booking removes its reminder.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

Screenshots or a screen recording across both phones, plus the Firestore result

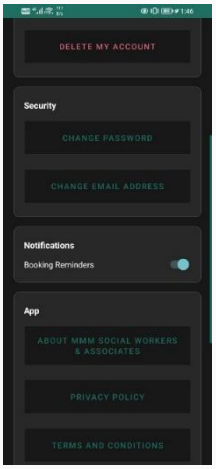


Figure INT-06

INT-07

Owner	All three members

Steps / what to check	Offline and recovery
Expected result	Go offline mid-use, then reconnect: banner appears and clears, nothing is duplicated, live lists catch up.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

Screenshots or a screen recording across both phones, plus the Firestore result

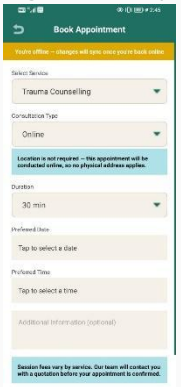


Figure INT-07

INT-08

Owner	All three members
Steps / what to check	Session persistence
Expected result	Log in, kill the app, reopen offline: user is routed to the correct home screen using the cached role.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

Screenshots or a screen recording across both phones, plus the Firestore result

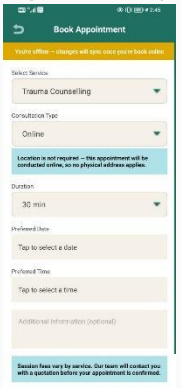


Figure INT-08

INT-09

Owner	All three members
Steps / what to check	Analytics and export

Expected result	Admin Analytics numbers match what you count by hand in Firestore; CSV export contains the same bookings.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

Screenshots or a screen recording across both phones, plus the Firestore result



Figure INT-09

INT-10

Owner	All three members
Steps / what to check	Two clients, one slot
Expected result	Clients A and B both try the final free slot at the same time: exactly one booking exists, both users see a correct result.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

Screenshots or a screen recording across both phones, plus the Firestore result

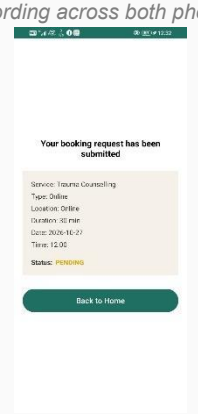


Figure INT-10

INT-11

Owner	All three members
Steps / what to check	Search and filter
Expected result	Admin dashboard search and filters return the right bookings and update live when a new booking arrives.

Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

Screenshots or a screen recording across both phones, plus the Firestore result



Figure INT-11

INT-12

Owner	All three members
Steps / what to check	Account changes
Expected result	Change email and password, then log out and in with the new details; bookings are still linked to the same user.
Result	**PASS**
Device / date	Phone and Table Date: 15/10/2026

Screenshots or a screen recording across both phones, plus the Firestore result

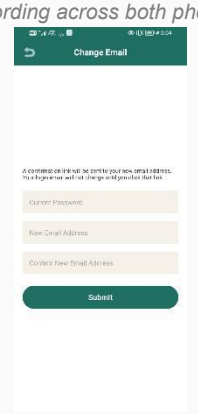


Figure INT-12

8. Defect log (B1 to B27)

B1 to B12 were found during development. B13 to B27 were found in the final implementation audit and fixed in the hardening phases.

#	Defect	Root cause	Resolution
B1	Admin Dashboard XML failed to compile after adding app:tint	Missing xmlns:app namespace on the root layout tag	Added the namespace declaration
B2	Logout unreachable on Settings	Buttons sat below the visible scroll area, no scroll indicator	Added a scroll thumb and repositioned logout
B3	Emulator 'Unknown error' on run	Emulator process had hung	Cold-booted the emulator
B4	PackageInfo import error	Wrong package path (android.pm vs android.content.pm)	Corrected the import
B5	Test files could not resolve JUnit/Espresso imports	Files placed in src/main/java instead of src/test / src/androidTest	Moved to the correct source sets
B6	strings.xml failed to parse	A stray invalid closing tag was accidentally included	Removed the invalid tag
B7	Admin login crashed after the date-filter redesign	tvFilterDate changed to a LinearLayout in XML but the Java field type was not updated	Changed the field type to View
B8	Register consent text clipped	Checkbox row height/text-wrap not configured	wrap_content rows, maxLines/ellipsize none, explicit bottom margins
B9	Status badges hard to read in light mode	Black text on dark badge backgrounds	Bold white badge text for all four statuses
B10	Back on Home returned to Welcome (looked like a logout)	Welcome stayed on the back stack; CLEAR_TOP did not clear it	Double-press-to-exit; navigation flags changed to NEW_TASK | CLEAR_TASK
B11	Register did not centre vertically on tablet	NestedScrollView + fillViewport + weight-based centring does not reliably stretch	Plain ScrollView with a constrained content column
B12	Register header dragged to the middle on tablet	Header and form shared one centred container	Layout restructured so the header stays at the top
B13	A user could register themselves as an admin	users create rule checked ownership but not the role value	Rule now requires role == client on create
B14	A client could confirm their own booking or mark it paid	bookings update rule allowed the owner to change any field	Clients limited to status to cancelled; all else admin-only
B15	Any user could overwrite or delete any time slot	bookedSlots allowed any signed-in user to write	Slots carry clientId; create/delete limited to owner or admin; update denied
B16	Cancel/decline could leave a stuck slot	Status update and slot delete were two separate calls	One Firestore transaction (BookingActions)
B17	Bookings had no true Delete	Cancel only set status to cancelled	Admin Delete Booking (permanent, confirmed, atomic)
B18	Two different clients could book overlapping times	Overlap only checked the same client's bookings; slot IDs ignored duration	Per-15-minute block reservation inside a transaction
B19	An appointment could finish after closing time	Only the start hour was compared with closing time	Start + duration checked; re-checked on duration change and submit
B20	Admin's Operating Days setting had no effect	Booking screen hardcoded Saturday/Sunday	Structured openDays list with checkboxes, enforced when booking
B21	Booking Reminders switch did nothing	Preference saved but never read before notifying	Enforced in NotificationHelper; reminders cancelled/rescheduled on toggle

#	Defect	Root cause	Resolution
B22	Failed registration could leave an orphaned login	Auth account created, then profile write failed with no rollback	Auth account deleted if the profile write fails
B23	Booking with no status could crash screens	Unguarded switch on a null status	StatusUtils treats missing status as pending
B24	Holidays only checked for the current year	Holidays fetched once, for the current year	Fetched and cached per year
B25	Email validation accepted '@' and 'a@b'	Rule was contains('@')	Regex for local part, @ and dotted domain; tests updated
B26	Default test stubs and .idea/ committed	Never removed	Deleted; .idea untracked and ignored
B27	Opening Settings could show a false 'reminders enabled' toast	Switch listener attached before the saved value loaded	Listener attached only after the saved value is applied

Tests and Results

MMM Booking App

Appointment booking for MMM Social Workers and Associates (Android, Java, Firebase)

Group member	Student number	Testing responsibility
Faith Murau	EDUV4821116	UX/UI testing
Khauhelo Moqechane	MD.2022.Q7M4P1	Screen and navigation testing
Ignatius Mayibongwe Khumalo	EDUV4960805	Database, authentication and booking logic testing
All three members		Integration testing, GitHub, demonstration video, final report

Links

Item	Link
GitHub repository	https://github.com/IgnatiusKhumalo/MMM-Booking-App.git
Testing evidence (screenshots, by test ID)	Testing_Evidence_MMM_Booking_App-Group11.pdf in the repository root
Test record (every test)	TESTING.md in the repository root
Demonstration video	Group11_MMM-BOOKING_APP 2026-10-23 at 8.47.38 PM.mp4 in the repository

Where each marking criterion is answered

Rubric criterion (marks)	Section
Definition of testing objectives and scope (5)	1. Objectives and scope; 2. Testing plan
Core functionality tested with visible results and clear outcomes (15)	4.1 Functional testing; Appendix A (evidence)
UX checks and failure / invalid input scenarios handled (10)	4.2 Usability testing; 4.3 Edge-case and invalid-input testing
Well-structured table or log showing outcomes and notes (10)	3. Summary; 4. Detailed results (every test: expected, actual, status, notes)
System runs without crashing; issues identified and resolved (10)	5. Bugs found and fixed; 6. Stability

1. Objectives and scope

1.1 Purpose

This document reports the testing of the MMM Booking App, a native Android application (Java) that lets clients of MMM Social Workers and Associates register, book and cancel appointments, and lets practice administrators manage bookings and practice information. The back end is Firebase Authentication and Cloud Firestore, with the Nager.Date public holiday API and local notifications. All tests were run on the real working application.

1.2 Testing objectives

- Confirm that every implemented feature works as specified (functional testing), including login and logout, create, read, update and delete (CRUD) operations, data storage and retrieval, and screen navigation.
- Evaluate the user experience: layout, consistency, clarity of labels and messages, touch targets, dark mode and tablet layout (usability testing).
- Confirm that the app handles blank forms, invalid or boundary input, offline use and abnormal data without crashing, and gives clear feedback (edge-case testing).
- Confirm that data is secure: users cannot read other users' data, gain admin rights, tamper with bookings or overwrite other people's time slots (security and access control).
- Confirm that the booking rules prevent double booking, bookings outside opening hours or on holidays, and leave no stuck time slots (booking logic).
- Find, record, fix and retest defects, and show that the final app runs without crashing.

1.3 Scope

In scope	Out of scope
Registration, email verification, login, logout, forgot, change password and change email Client features: Home, Services, Book Appointment, Booking Confirmation, My Bookings, Contact, Settings, information screens Admin features: Dashboard, search and filters, Booking Detail (confirm, decline, complete, paid, delete), Practice Information, Analytics and CSV export Firestore data and security rules; booking slot logic; holiday API; local notifications and reminders Navigation and back stack; offline behaviour; dark mode; phone and tablet layouts	Load and performance testing with many concurrent users Server-sent push notifications (the app uses local notifications only) Client-side rescheduling (not implemented; clients cancel and rebook) Penetration testing beyond Firestore Security Rules checks Non-Android platforms (iOS, web)

1.4 Test environment

Item	Detail
Also used	Android Studio emulator profiles during development; second phone or emulator for two-user tests
Back end	Firebase project (Authentication, Cloud Firestore, region africa-south1) with published security rules matching firestore.rules.md
Test accounts	Client A, Client B (verified email addresses) and one Admin account created in the Firebase Console
Tools	Android Studio and JUnit (unit tests), Espresso (UI test), Firebase Console and Rules Playground, adb (alarm check), device screen recorder
Test period	**From 15/10/2026 to 30/10/2026**

1.5 Pass and fail criteria

- PASS: the actual behaviour matches the expected result, with evidence recorded under the same test ID.
- FAIL: the behaviour differs, the app crashes, or data is wrong. Every failure is logged as a defect (section 5), fixed where possible, and retested.
- A test is marked PASS only after it has been run. A test that retests a defect is marked PASS only after the fix and a successful retest.

2. Testing plan

2.1 Approach

Testing combined automated and manual methods. Pure logic (input validation, 15-minute slot reservation, open days and hours, status handling) is covered by 61 automated JUnit tests and one Espresso UI test. Everything that depends on the device, Firebase or the interface was tested manually by following written test cases, each with an ID, steps and an expected result. Security rules were tested both through the app and by simulating requests in the Firebase Rules Playground, and each backend action was checked in the Firebase Console.

Tests were split by team member (UX/UI; screens and navigation; database, authentication and booking logic) with integration testing done by all three using two phones: one as a client and one as the administrator, so real-time updates could be observed.

2.2 What is tested and how

Component or feature	Test types	Test IDs
Authentication: register, verify, login, logout, reset and change password and email	Functional, edge case, security	F1 to F5, F19, F22, F24, F25, DB-01 to DB-13
Booking creation, validation and conflict prevention	Functional, edge case	F6 to F9, BK-01 to BK-22, PC1 to PC8
Cancellation, decline and delete (CRUD)	Functional	F9, F10, PB1 to PB5, BK-15 to BK-20
Admin features: dashboard, search, export, practice information, analytics	Functional	F10 to F15, DB-17 to DB-20
Data storage and retrieval (Firestore documents)	Functional	DB-14 to DB-21
Security rules and access control	Functional, security	PA1 to PA3, PM-01 to PM-18
Screen transitions, navigation and back stack	Functional, usability	NAV-01 to NAV-18, F21, F22, screen inventory
User input validation (UI) and error messages	Usability, edge case	UX-15 to UX-24, E1 to E8, PF3 to PF5
UI responsiveness, layout, dark mode, tablet	Usability	U1 to U12, UX-01 to UX-14
Notifications and reminders	Functional	PD1 to PD7, F18, INT-06
Offline and robustness	Edge case	PE1 to PE5, E2 to E5, BK-21, NAV-18, INT-07, INT-08
Cross-component flows (UI, Firebase, API, notifications)	Integration	INT-01 to INT-12
Business logic and validation helpers	Automated unit tests	ValidationUtilsTest, BookingRulesTest, Espresso Welcome test

A test can cover more than one component, so IDs appear in more than one row. The complete list of tests is in section 4.

2.3 Test case format

Every test has: an ID, the feature, what was tested (steps), the expected result, the actual result, a PASS / FAIL status, and notes (evidence reference, defect reference or improvement). Evidence for each ID (a screenshot, or a screen recording for real-time behaviour) is stored under the same ID in the testing evidence document.

2.4 Test data and conditions

- Valid, invalid and boundary values: names with one word, phone numbers with 8 and 10 digits, passwords missing each required character class, emails such as abc@ and a@b.
- Dates: a closed day, a South African public holiday (including one in the following year), and normal working days.
- Times: before opening, after closing, appointments that run past closing, and start times off the quarter hour.
- Conditions: online and offline (Airplane Mode), two users acting at the same time, portrait and landscape, phone and tablet, light and dark mode.

3. Test results summary

All 258 tests were executed and passed. In addition, one Espresso UI test confirms that the Welcome screen shows the Login and Register buttons. Several tests were written to retest defects found earlier (section 5); those were marked PASS only after the fix was in place and the test was repeated.

Test group	Total	Pass	Fail	Pending
Automated unit tests (ValidationUtilsTest 37, BookingRulesTest 24)	61	61	0	0
Hardening verification (PA1 to PF9)	37	37	0	0
Functional (F1 to F25)	25	25	0	0
Usability (U1 to U12)	12	12	0	0
Edge cases (E1 to E8)	8	8	0	0
Subtotal: core test plan	143	143	0	0
Team: UX/UI (UX-01 to UX-24)	24	24	0	0
Team: navigation (NAV-01 to NAV-18)	18	18	0	0
Team: database and authentication (DB-01 to DB-21)	21	21	0	0
Team: booking logic (BK-01 to BK-22)	22	22	0	0
Team: permissions (PM-01 to PM-18)	18	18	0	0
Team: integration (INT-01 to INT-12)	12	12	0	0
Subtotal: team test plans	115	115	0	0
TOTAL	258	258	0	0

Key observations

- Three security defects (B13 to B15) were found by the access-control tests: a user could register as admin, a client could confirm or mark their own booking as paid, and any user could overwrite another user's time slot. All three were fixed in the Firestore rules and retested.
- The booking-logic tests found that overlapping bookings by different clients were possible (B18) and that an appointment could finish after closing time (B19). Both were fixed with a transaction that reserves every 15-minute block.
- Cancel and decline could leave a stuck time slot (B16); they now use one atomic operation.
- Two crash defects (B7, B23) were found and fixed. No crash occurred in the final test run.

4. Detailed test results

Each row is one test. Actual result "As expected." means the observed behaviour matched the expected result exactly. Rows that verify a defect fix say "after the fix for" and point to the defect number in section 5. Evidence for every row is under the same ID in the testing evidence document.

4.1 Functional testing

4.1.1 Core features (F1 to F25)

ID	Feature	What was tested	Expected result	Actual result	Status	Notes / evidence
F1	Registration	Fill all fields correctly, tick consent, submit	Account created, verification email sent	As expected.	**PASS**	Evidence: F1, PA1
F2	Registration blocked without consent	Fill all fields, leave consent unticked, submit	Red error shown, no account created	As expected.	**PASS**	Evidence: F2
F3	Email verification gate	Log in before verifying email	Blocked, resend option shown	As expected.	**PASS**	Evidence: F3
F4	Login (verified account)	Log in with correct credentials	Routed to Home (client) or Admin Dashboard (admin), with a brief "Logged in successfully" toast	As expected.	**PASS**	Evidence: F4
F5	Password reset	Request reset for a registered email	Reset email received	As expected.	**PASS**	Evidence: F5
F6	Booking submission	Complete and submit a valid booking	Confirmation screen shown, booking appears in My Bookings, immediate notification received	As expected.	**PASS**	Evidence: F6, PD1
F7	Double-booking prevention	Two accounts attempt overlapping times	Second attempt rejected	As expected, after the fix for B18.	**PASS**	Evidence: F7, PC3. Retest of defect B18.
F8	Public holiday block	Attempt a known SA public holiday date	Booking date rejected with a clear message	As expected, after the fix for B24.	**PASS**	Evidence: F8, PC8. Retest of defect B24.
F9	Cancellation	Cancel a pending/confirmed booking	Confirmation dialog shown, status becomes Cancelled, slot freed	As expected, after the fix for B16.	**PASS**	Evidence: F9, PB1. Retest of defect B16.
F10	Admin confirm/decline	Admin confirms a pending booking	Status updates automatically on the client's My Bookings screen in real time	As expected.	**PASS**	Evidence: F10
F11	Admin search	Type a service name into the search box	List filters to matching bookings only	As expected.	**PASS**	Evidence: F11

ID	Feature	What was tested	Expected result	Actual result	Status	Notes / evidence
F12	Admin search + filters combined	Apply a status filter, then search	Only bookings matching both are shown	As expected.	**PASS**	Evidence: F12
F13	CSV export	Tap Export CSV on Admin Dashboard	Share sheet opens with a valid CSV file	As expected.	**PASS**	Evidence: F13
F14	Practice Information edit	Admin changes open days/phone, saves	Client's Contact and booking screens reflect the change	As expected, after the fix for B20.	**PASS**	Evidence: F14, PC6. Retest of defect B20.
F15	Admin Analytics	Open View Detailed Analytics	Counts, % change and bar charts display for both Bookings Created and New Registrations	As expected.	**PASS**	Evidence: F15
F16	Settings — single logout	Open Settings, scroll to Support	Exactly one Logout button present	As expected.	**PASS**	Evidence: F16
F17	Dark Mode toggle confirmation	Toggle Dark Mode on Settings	Toast confirms enabled/disabled; theme changes across the app	As expected.	**PASS**	Evidence: F17, PD7
F18	Booking Reminders toggle	Toggle Booking Reminders on Settings	Toast confirms enabled/disabled; notifications obey the switch	As expected, after the fix for B21.	**PASS**	Evidence: F18. Retest of defect B21.
F19	Logout confirmation dialog	Tap Logout from Home, Settings, and Admin Dashboard	Cancel/Log Out dialog each time; Cancel does nothing; Log Out returns to Welcome	As expected, after the fix for B2.	**PASS**	Evidence: F19. Retest of defect B2.
F20	Delete Account explainer	Tap Delete My Account on Settings	Explainer first; Continue opens the pre-addressed email; Cancel dismisses	As expected.	**PASS**	Evidence: F20
F21	Back button on Home/Admin Dashboard	Press back once, then again within 2 seconds	First press shows "Press back again to exit"; second exits; user stays logged in	As expected, after the fix for B10.	**PASS**	Evidence: F21, PE4. Retest of defect B10.
F22	Session persistence	Log in, fully close the app, reopen	Opens directly to Home/Admin Dashboard without logging in again	As expected.	**PASS**	Evidence: F22, PE3
F23	Real-time admin list update	With Admin Dashboard open, a client submits a booking from another device	New booking and updated counts appear automatically	As expected.	**PASS**	Evidence: F23
F24	Confirm-email mismatch on Register	Type different addresses in Email and Confirm Email, submit	"Email addresses do not match" shown; no account created	As expected.	**PASS**	Evidence: F24
F25	Change Email	Settings → Change Email with current password and a new address	"Confirmation Sent" dialog; link arrives at the new address; login email unchanged until clicked	As expected.	**PASS**	Evidence: F25

4.1.2 Screens and navigation (screen inventory and NAV-01 to NAV-18)

Screen inventory: every screen was opened once from the screen it is reached from.

Screen	Reached from	Notes	Opened
Splash / router	App launch	Routes to Welcome, client Home or admin Dashboard depending on session	**PASS**
Welcome	Splash (logged out)	Opens with working Login and Register buttons.	**PASS**
Login	Welcome	Opens. Login works for both client and admin.	**PASS**
Register	Welcome	Opens. Form and validation errors show.	**PASS**
Forgot Password	Login	Opens. Reset email request works.	**PASS**
Home	Login (client)	Opens after client login.	**PASS**
Services	Home	Opens and lists the services.	**PASS**
Book Appointment	Home or Services	Opens. Booking form works.	**PASS**
Booking Confirmation	After a successful booking	Opens after submitting a valid booking.	**PASS**
My Bookings	Home	Opens and lists the client's bookings.	**PASS**
Contact	Home	Opens.	**PASS**
Settings	Home or menu	Opens. Dark mode and reminders available.	**PASS**
Change Password / Change Email	Settings	Both open from Settings.	**PASS**
FAQ / About / Privacy / Terms	Settings or Register (terms)	All four open.	**PASS**
Admin Dashboard	Login (admin)	Opens after admin login, not the client Home.	**PASS**
Booking Detail	Dashboard list item	Opens when a booking is tapped.	**PASS**
Practice Information	Dashboard	Opens.	**PASS**
Admin Analytics	Dashboard	Opens.	**PASS**

Navigation and back-stack tests:

ID	Feature	What was tested	Expected result	Actual result	Status	Notes / evidence
NAV-01	Navigation	Fresh install, open app	Welcome screen with working Login and Register buttons.	As expected.	**PASS**	Evidence: NAV-01
NAV-02	Navigation	Login with a verified client	Lands on Home. Pressing back does not return to Login.	As expected, after the fix for B10.	**PASS**	Evidence: NAV-02. Retest of defect B10.

ID	Feature	What was tested	Expected result	Actual result	Status	Notes / evidence
NAV-03	Navigation	Login with admin	Lands on Admin Dashboard, not client Home.	As expected.	**PASS**	Evidence: NAV-03
NAV-04	Navigation	Close app (swipe away) while logged in, reopen	Goes straight to the correct home screen for the role, no login required.	As expected.	**PASS**	Evidence: NAV-04
NAV-05	Navigation	Log out, press back	Cannot return to a logged-in screen.	As expected, after the fix for B10.	**PASS**	Evidence: NAV-05. Retest of defect B10.
NAV-06	Navigation	Welcome to Register to Login using on-screen links, then back	Back moves one step at a time with no crash or blank screen.	As expected.	**PASS**	Evidence: NAV-06
NAV-07	Navigation	Register successfully	Verification message shown, user returned to a sensible screen (Login), not left in a half-logged-in state.	As expected.	**PASS**	Evidence: NAV-07
NAV-08	Navigation	Login, then Forgot Password, then back	Returns to Login with fields intact.	As expected.	**PASS**	Evidence: NAV-08
NAV-09	Navigation	Home, then Book Appointment, then submit	Booking Confirmation opens; back from Confirmation goes Home or My Bookings, not back into the filled booking form (the form is closed on submit).	As expected.	**PASS**	Evidence: NAV-09
NAV-10	Navigation	Home, then My Bookings, cancel a booking, then back	List already updated (live listener), back goes Home.	As expected.	**PASS**	Evidence: NAV-10
NAV-11	Navigation	Settings, then Change Password, back; Change Email, back	Returns to Settings each time.	As expected.	**PASS**	Evidence: NAV-11
NAV-12	Navigation	Terms/Privacy opened from Register, then back	Registration form keeps its typed values.	As expected.	**PASS**	Evidence: NAV-12
NAV-13	Navigation	Admin Dashboard, then Booking Detail, confirm, back	Dashboard shows the updated status.	As expected.	**PASS**	Evidence: NAV-13
NAV-14	Navigation	Admin Dashboard: press back once, then again quickly	First press shows an exit hint, second exits (double-press behaviour).	As expected, after the fix for B10.	**PASS**	Evidence: NAV-14. Retest of defect B10.
NAV-15	Navigation	Rotate device on every major screen	No crash; data preserved or sensibly reloaded.	As expected.	**PASS**	Evidence: NAV-15
NAV-16	Navigation	Rapidly tap a button twice (Login, Submit Booking)	Only one action happens (button disabled while loading); no duplicate booking.	As expected.	**PASS**	Evidence: NAV-16

ID	Feature	What was tested	Expected result	Actual result	Status	Notes / evidence
NAV-17	Navigation	Open app from a notification (reminder or booking submitted)	Opens the app without a crash.	As expected.	**PASS**	Evidence: NAV-17
NAV-18	Navigation	Turn off data, then tap navigation items	Screens open; the offline banner shows; no crash.	As expected.	**PASS**	Evidence: NAV-18

Result of tapping every button, icon, list item and link: all 18 screens opened and no button that did nothing, crashed or went to the wrong place was found. Delete Account in Settings shows an explainer and then opens a pre-addressed email to the practice (see F20), which is a documented design decision.

4.1.3 Authentication (DB-01 to DB-13)

ID	Feature	What was tested	Expected result	Actual result	Status	Notes / evidence
DB-01	Authentication	Register Client A with valid details	Firebase Authentication shows the new user; users/{uid} exists with name, email, phone and role: "client"; verification email arrives.	As expected.	**PASS**	Evidence: DB-01
DB-02	Authentication	Login before verifying email	Blocked, with an option to resend the verification email.	As expected.	**PASS**	Evidence: DB-02
DB-03	Authentication	Verify via email link, log in	Succeeds and routes to Home.	As expected.	**PASS**	Evidence: DB-03
DB-04	Authentication	Wrong password; unregistered email	Login fails with a message; no session created.	As expected.	**PASS**	Evidence: DB-04
DB-05	Authentication	Register with an already used email	Clear failure, no duplicate user document.	As expected.	**PASS**	Evidence: DB-05
DB-06	Authentication	Boundary data: password with 7 chars, no uppercase, no digit, no special; phone 8 and 10 digits; name "A B"	Each is rejected.	As expected.	**PASS**	Evidence: DB-06
DB-07	Authentication	Email without a proper domain, for example abc@ or a@b	Rejected with the email error (the email format is checked, not just the @ sign).	As expected, after the fix for B25.	**PASS**	Evidence: DB-07. Retest of defect B25.
DB-08	Authentication	Forgot Password with a registered email	Reset email received and the new password works.	As expected.	**PASS**	Evidence: DB-08
DB-09	Authentication	Change Password with correct old password; with wrong old password	Succeeds / rejected. Login with the new password works.	As expected.	**PASS**	Evidence: DB-09
DB-10	Authentication	Change Email	Verification flow works and the users document email stays consistent.	As expected.	**PASS**	Evidence: DB-10
DB-11	Authentication	Logout, then relaunch	Back at Welcome, no session.	As expected.	**PASS**	Evidence: DB-11
DB-12	Authentication	Register while offline	Fails with a message and no orphaned account is left in Firebase Authentication.	As expected, after the fix for B22.	**PASS**	Evidence: DB-12. Retest of defect B22.
DB-13	Authentication	Login as the admin account	Routed to the admin dashboard; client accounts never reach it.	As expected.	**PASS**	Evidence: DB-13

4.1.4 Data storage and retrieval (DB-14 to DB-21)

ID	Feature	What was tested	Expected result	Actual result	Status	Notes / evidence
DB-14	Data storage and retrieval	Client A books In-Person, 60 min, with notes	bookings/{id} has clientId = A's uid, service, consultationType in-person, location, duration 60, date yyyy-MM-dd, time HH:mm, additionalInfo, status pending, paid false, createdAt.	As expected.	**PASS**	Evidence: DB-14
DB-15	Data storage and retrieval	Same booking in My Bookings	All details match what was entered.	As expected.	**PASS**	Evidence: DB-15
DB-16	Data storage and retrieval	Book Online	location is null and consultationType is online.	As expected.	**PASS**	Evidence: DB-16
DB-17	Data storage and retrieval	Admin opens Booking Detail	Shows client name, phone and email read from users, plus all booking fields.	As expected.	**PASS**	Evidence: DB-17
DB-18	Data storage and retrieval	Admin confirms, marks paid, marks completed	Firestore status and paid change each time and Client A sees the new status live without refreshing.	As expected.	**PASS**	Evidence: DB-18
DB-19	Data storage and retrieval	Edit Practice Information (days, hours)	practiceInfo document updates and the Book Appointment screen immediately uses the new hours.	As expected, after the fix for B20.	**PASS**	Evidence: DB-19. Retest of defect B20.
DB-20	Data storage and retrieval	Practice hours validation: end before start; malformed times; no open days	Each rejected with a message; nothing saved.	As expected, after the fix for B20.	**PASS**	Evidence: DB-20. Retest of defect B20.
DB-21	Data storage and retrieval	Settings: toggle reminders and dark mode	Preference saved on the account/device and still applied after restart.	As expected.	**PASS**	Evidence: DB-21

4.1.5 Booking logic (BK-01 to BK-22)

ID	Feature	What was tested	Expected result	Actual result	Status	Notes / evidence
BK-01	Booking logic	Book 10:00 for 60 min	Four slot documents appear with Client A's uid as clientId.	As expected.	**PASS**	Evidence: BK-01
BK-02	Booking logic	Client B books the same location, same date, 10:00	Rejected: time already reserved. No booking or slots created for B.	As expected.	**PASS**	Evidence: BK-02
BK-03	Booking logic	Client B books 10:30, 30 min (overlaps the middle of A's)	Rejected. This is the key overlap test; the original version of the app wrongly allowed it.	As expected, after the fix for B18.	**PASS**	Evidence: BK-03. Retest of defect B18.
BK-04	Booking logic	Client B books 09:00 for 60 min (ends exactly at 10:00)	Accepted: back-to-back appointments are allowed.	As expected.	**PASS**	Evidence: BK-04
BK-05	Booking logic	Client B books 11:00	Accepted (A's last block is 10:45 to 11:00).	As expected.	**PASS**	Evidence: BK-05
BK-06	Booking logic	Client B books the same time at a different location	Accepted: each location is its own calendar.	As expected.	**PASS**	Evidence: BK-06
BK-07	Booking logic	Client A books another appointment overlapping her own	Rejected with the 'overlaps your own booking' message.	As expected.	**PASS**	Evidence: BK-07
BK-08	Booking logic	Two phones submit the same slot at the same moment (repeat three times)	Only one succeeds (transaction).	As expected, after the fix for B18.	**PASS**	Evidence: BK-08. Retest of defect B18.
BK-09	Booking logic	Book on a closed day (for example Sunday)	Rejected, message names the open days.	As expected.	**PASS**	Evidence: BK-09
BK-10	Booking logic	Book on a South African public holiday (for example 25 December, 1 January), and a holiday in next year	Rejected with the holiday message; the next-year date confirms the per-year fetch.	As expected, after the fix for B24.	**PASS**	Evidence: BK-10. Retest of defect B24.
BK-11	Booking logic	Book before opening time; at 16:45 for 60 min when closing is 17:00	Rejected with the correct reason (before opening; runs past closing).	As expected, after the fix for B19.	**PASS**	Evidence: BK-11. Retest of defect B19.
BK-12	Booking logic	Choose 16:00 with 30 min, then change duration to 60 min while closing is 17:00	If it no longer fits, the time is cleared and the user told.	As expected, after the fix for B19.	**PASS**	Evidence: BK-12. Retest of defect B19.
BK-13	Booking logic	Choose a time with minutes not on the quarter hour (10:07)	Rejected.	As expected.	**PASS**	Evidence: BK-13

ID	Feature	What was tested	Expected result	Actual result	Status	Notes / evidence
BK-14	Booking logic	Book with no date selected; with no time	Rejected with a message.	As expected.	**PASS**	Evidence: BK-14
BK-15	Booking logic	Cancel Client A's booking (confirm dialog)	Booking status becomes cancelled; all four slot documents are deleted in the same operation; reminder cancelled.	As expected, after the fix for B16.	**PASS**	Evidence: BK-15. Retest of defect B16.
BK-16	Booking logic	Client B immediately books the freed time	Accepted.	As expected.	**PASS**	Evidence: BK-16
BK-17	Booking logic	Cancel an already cancelled or completed booking	Cancel button not offered, or a clear 'cannot be cancelled' message.	As expected.	**PASS**	Evidence: BK-17
BK-18	Booking logic	Admin declines a pending booking (confirm dialog)	Same result as BK-15 and the record is kept as cancelled.	As expected, after the fix for B16.	**PASS**	Evidence: BK-18. Retest of defect B16.
BK-19	Booking logic	Admin deletes a booking (confirm dialog)	Booking document is gone; slots are freed if it was still active; Client A's list updates.	As expected, after the fix for B17.	**PASS**	Evidence: BK-19. Retest of defect B17.
BK-20	Booking logic	Delete an old cancelled booking whose time has since been re-booked by someone else	The new booking's slots are not removed.	As expected, after the fix for B16.	**PASS**	Evidence: BK-20. Retest of defect B16.
BK-21	Booking logic	Go offline, then submit a booking and cancel a booking	Both blocked with the offline message; no partial data in Firestore.	As expected.	**PASS**	Evidence: BK-21
BK-22	Booking logic	Book an Online appointment at 10:00, then another client books Online at 10:00	Rejected, because all online bookings share one calendar. Record this as a design limitation (one online appointment at a time).	As expected.	**PASS**	Evidence: BK-22

4.1.6 Database permissions and access control (PM-01 to PM-18)

ID	Feature	What was tested	Expected result	Actual result	Status	Notes / evidence
PM-01	Access control (security rules)	Unauthenticated read of bookings or users	Denied.	As expected.	**PASS**	Evidence: PM-01
PM-02	Access control (security rules)	Unauthenticated read of practiceInfo	Allowed (public by design, needed for the app before login).	As expected.	**PASS**	Evidence: PM-02
PM-03	Access control (security rules)	Client A reads users/{A}	Allowed.	As expected.	**PASS**	Evidence: PM-03
PM-04	Access control (security rules)	Client A reads users/{B}	Denied.	As expected.	**PASS**	Evidence: PM-04
PM-05	Access control (security rules)	Admin reads any users document	Allowed.	As expected.	**PASS**	Evidence: PM-05
PM-06	Access control (security rules)	Create users/{A} with role: "admin"	Denied (create requires role client). This was a privilege-escalation hole in the first version.	As expected, after the fix for B13.	**PASS**	Evidence: PM-06. Retest of defect B13.
PM-07	Access control (security rules)	Client A updates own profile and changes role to admin	Denied.	As expected.	**PASS**	Evidence: PM-07
PM-08	Access control (security rules)	Client A reads own booking	Allowed.	As expected.	**PASS**	Evidence: PM-08
PM-09	Access control (security rules)	Client A reads Client B's booking	Denied.	As expected.	**PASS**	Evidence: PM-09
PM-10	Access control (security rules)	Client A creates a booking with status confirmed, or paid true, or another user's clientId	Denied.	As expected, after the fix for B14.	**PASS**	Evidence: PM-10. Retest of defect B14.
PM-11	Access control (security rules)	Client A updates own booking to status cancelled only	Allowed.	As expected.	**PASS**	Evidence: PM-11
PM-12	Access control (security rules)	Client A updates own booking to confirmed, or sets paid: true, or changes the date	Denied.	As expected, after the fix for B14.	**PASS**	Evidence: PM-12. Retest of defect B14.
PM-13	Access control (security rules)	Client A deletes a booking	Denied.	As expected.	**PASS**	Evidence: PM-13

ID	Feature	What was tested	Expected result	Actual result	Status	Notes / evidence
PM-14	Access control (security rules)	Admin updates status and paid; admin deletes a booking	Allowed.	As expected.	**PASS**	Evidence: PM-14
PM-15	Access control (security rules)	Client A creates a slot with clientId = B's uid	Denied.	As expected, after the fix for B15.	**PASS**	Evidence: PM-15. Retest of defect B15.
PM-16	Access control (security rules)	Client A deletes Client B's slot	Denied.	As expected, after the fix for B15.	**PASS**	Evidence: PM-16. Retest of defect B15.
PM-17	Access control (security rules)	Anyone updates (modifies) a slot	Denied.	As expected, after the fix for B15.	**PASS**	Evidence: PM-17. Retest of defect B15.
PM-18	Access control (security rules)	Client A writes practiceInfo	Denied. Admin write allowed.	As expected.	**PASS**	Evidence: PM-18

4.1.7 Hardening verification (PA1 to PF9)

ID	Feature	What was tested	Expected result	Actual result	Status	Notes / evidence
PA1	Security rules	Register a brand-new test account	Account created normally; role defaults to client	As expected, after the fix for B13.	**PASS**	Evidence: PA1. Retest of defect B13.
PA2	Security rules	Confirm a client cannot set status/paid on a booking (Firebase Rules Playground)	The write is denied by the rules	As expected, after the fix for B14.	**PASS**	Evidence: PA2. Retest of defect B14.
PA3	Security rules	Register → book → cancel as a client	Whole flow still works under the new rules	As expected, after the fix for B14.	**PASS**	Evidence: PA3. Retest of defect B14.
PB1	Booking CRUD	Client cancels a pending booking	Shows Cancelled; the same time slot can be booked again	As expected, after the fix for B16.	**PASS**	Evidence: PB1. Retest of defect B16.
PB2	Booking CRUD	Admin declines a pending booking	Confirmation dialog first; then Cancelled and slot freed	As expected, after the fix for B16.	**PASS**	Evidence: PB2. Retest of defect B16.
PB3	Booking CRUD	Admin deletes a booking	Confirmation first; screen closes; booking disappears from the dashboard; the bookings document is gone in the Firebase Console	As expected, after the fix for B17.	**PASS**	Evidence: PB3. Retest of defect B17.
PB4	Booking CRUD	Admin deletes a cancelled booking whose slot was since re-booked by another client	The other client's slot document still exists	As expected, after the fix for B16.	**PASS**	Evidence: PB4. Retest of defect B16.
PB5	Booking CRUD	Client has no delete action	No client delete button; the server-side rule is the protection	As expected.	**PASS**	Evidence: PB5
PC1	Booking rules	Closing time (hours 08:00–18:00)	60 min at 17:45 rejected; 17:00 accepted; 17:30 + 60 min rejected, switching to 30 min fits	As expected, after the fix for B19.	**PASS**	Evidence: PC1. Retest of defect B19.
PC2	Booking rules	Quarter-hour starts	10:20 rejected with the quarter-hour message; 10:15 accepted	As expected.	**PASS**	Evidence: PC2
PC3	Booking rules	Cross-client overlap	Client A 10:00 (60 min); Client B at 10:30 and at 10:00 both rejected; 11:00 accepted	As expected, after the fix for B18.	**PASS**	Evidence: PC3. Retest of defect B18.
PC4	Booking rules	Same-client overlap	Client A booking 10:30 while holding 10:00–11:00 gets the "you already have a booking that overlaps" message	As expected, after the fix for B18.	**PASS**	Evidence: PC4. Retest of defect B18.

ID	Feature	What was tested	Expected result	Actual result	Status	Notes / evidence
PC5	Booking rules	Slot freed on cancel	After A cancels, B can book 10:00 straight away	As expected, after the fix for B16.	**PASS**	Evidence: PC5. Retest of defect B16.
PC6	Booking rules	Operating days	Untick Wednesday as admin → client cannot pick a Wednesday; Contact shows "Monday, Tuesday, Thursday - Friday"; restore afterwards	As expected, after the fix for B20.	**PASS**	Evidence: PC6. Retest of defect B20.
PC7	Booking rules	Admin hours validation	8:00, 25:00 and opening-after-closing each rejected; saving with no days ticked rejected	As expected, after the fix for B20.	**PASS**	Evidence: PC7. Retest of defect B20.
PC8	Booking rules	Future-year holiday	A date in next January is checked without error; 1 January is rejected	As expected, after the fix for B24.	**PASS**	Evidence: PC8. Retest of defect B24.
PD1	Booking reminders	Switch ON (default): book an appointment	"Booking Submitted" notification appears	As expected, after the fix for B21.	**PASS**	Evidence: PD1. Retest of defect B21.
PD2	Booking reminders	Switch OFF, then book	"Disabled" toast; no notification; booking still succeeds	As expected, after the fix for B21.	**PASS**	Evidence: PD2. Retest of defect B21.
PD3	Booking reminders	Re-open Settings several times	No toast on opening — a toast only appears when the switch is tapped	As expected, after the fix for B27.	**PASS**	Evidence: PD3. Retest of defect B27.
PD4	Booking reminders	Scheduled reminder cancelled when OFF, restored when ON (adb shell dumpsys alarm)	Alarm present when ON, gone when OFF, back when ON again	As expected, after the fix for B21.	**PASS**	Evidence: PD4. Retest of defect B21.
PD5	Booking reminders	Persistence	Still OFF after fully closing the app; notificationsEnabled is false in the Console	As expected, after the fix for B21.	**PASS**	Evidence: PD5. Retest of defect B21.
PD6	Booking reminders	Account isolation	A second account on the same phone shows the switch ON	As expected, after the fix for B21.	**PASS**	Evidence: PD6. Retest of defect B21.
PD7	Booking reminders	Dark Mode toggle	Still works as before	As expected.	**PASS**	Evidence: PD7
PE1	Robustness	Registration rollback (temporarily block profile creation in the rules)	"Registration failed" toast; the new email is not left in Authentication → Users	As expected, after the fix for B22.	**PASS**	Evidence: PE1. Retest of defect B22.
PE2	Robustness	Delete a booking's status field in the Console	Admin Dashboard does not crash; booking shown/counted as Pending; Booking Detail opens	As expected, after the fix for B23.	**PASS**	Evidence: PE2. Retest of defect B23.
PE3	Robustness	Reopen the app in Airplane Mode (client and admin)	Lands on Home / Admin Dashboard, not Welcome	As expected.	**PASS**	Evidence: PE3

ID	Feature	What was tested	Expected result	Actual result	Status	Notes / evidence
PE4	Robustness	Regression: login routing and back button	Client → Home, admin → Dashboard with the "Logged in successfully" toast; double-back-to-exit works	As expected.	**PASS**	Evidence: PE4
PE5	Robustness	Logcat check	No email address in any LoginActivity/RegisterActivity log line	As expected.	**PASS**	Evidence: PE5
PF1	Clean-up and validation	Clean build after deleting the default test stubs	Compiles with no errors	As expected, after the fix for B26.	**PASS**	Evidence: PF1. Retest of defect B26.
PF2	Clean-up and validation	Run all unit tests	61 tests, all green	As expected.	**PASS**	Evidence: PF2
PF3	Clean-up and validation	Register with email thandi@example	Email error shown; thandi@example.com accepted	As expected, after the fix for B25.	**PASS**	Evidence: PF3. Retest of defect B25.
PF4	Clean-up and validation	Register with a weak password	Rules line turns red and bold, no toast; returns to normal when typing, green when valid	As expected.	**PASS**	Evidence: PF4
PF5	Clean-up and validation	Log In with empty fields	"This field is required" on each empty field	As expected.	**PASS**	Evidence: PF5
PF6	Clean-up and validation	Normal login	Client → Home, admin → Dashboard with the success toast	As expected.	**PASS**	Evidence: PF6
PF7	Clean-up and validation	Booking Detail as admin	Badge colour and Confirm/Decline/Complete/Delete behave as before; confirmation dialogs appear	As expected.	**PASS**	Evidence: PF7
PF8	Clean-up and validation	Comment-only files (Home, Change Email, Admin Analytics)	Behave exactly as before	As expected.	**PASS**	Evidence: PF8
PF9	Clean-up and validation	git status after untracking .idea	.idea no longer tracked	As expected, after the fix for B26.	**PASS**	Evidence: PF9. Retest of defect B26.

4.1.8 Integration testing (INT-01 to INT-12)

ID	Feature	What was tested	Expected result	Actual result	Status	Notes / evidence
INT-01	Integration	New client lifecycle	Register, verify email, login, view services, book, see confirmation, see booking in My Bookings, Firestore documents correct.	As expected.	**PASS**	Evidence: INT-01

ID	Feature	What was tested	Expected result	Actual result	Status	Notes / evidence
INT-02	Integration	Admin approval loop	Client books; admin dashboard updates in real time; admin confirms; client's My Bookings changes to confirmed without a refresh; admin marks paid, then completed.	As expected.	**PASS**	Evidence: INT-02
INT-03	Integration	Cancellation loop	Client cancels; admin dashboard shows cancelled; slots freed; second client can book the same time.	As expected.	**PASS**	Evidence: INT-03
INT-04	Integration	Admin changes practice hours	Admin changes closing time; client booking screen enforces it on the next attempt.	As expected.	**PASS**	Evidence: INT-04
INT-05	Integration	Holiday API integration	Booking blocked on a public holiday; then block the network and confirm the app warns that holidays could not be loaded and still enforces other rules.	As expected.	**PASS**	Evidence: INT-05
INT-06	Integration	Notifications	Booking submitted shows an immediate notification; the reminder for the evening before is scheduled; turning reminders off in Settings cancels them; turning on reschedules; cancelling a booking removes its reminder.	As expected.	**PASS**	Evidence: INT-06
INT-07	Integration	Offline and recovery	Go offline mid-use, then reconnect: banner appears and clears, nothing is duplicated, live lists catch up.	As expected.	**PASS**	Evidence: INT-07
INT-08	Integration	Session persistence	Log in, kill the app, reopen offline: user is routed to the correct home screen using the cached role.	As expected.	**PASS**	Evidence: INT-08
INT-09	Integration	Analytics and export	Admin Analytics numbers match what you count by hand in Firestore; CSV export contains the same bookings.	As expected.	**PASS**	Evidence: INT-09
INT-10	Integration	Two clients, one slot	Clients A and B both try the final free slot at the same time: exactly one booking exists, both users see a correct result.	As expected.	**PASS**	Evidence: INT-10

ID	Feature	What was tested	Expected result	Actual result	Status	Notes / evidence
INT-11	Integration	Search and filter	Admin dashboard search and filters return the right bookings and update live when a new booking arrives.	As expected.	**PASS**	Evidence: INT-11
INT-12	Integration	Account changes	Change email and password, then log out and in with the new details; bookings are still linked to the same user.	As expected.	**PASS**	Evidence: INT-12

.1.9 Automated unit tests (61) and Espresso UI test

Suite	Area	Tests	What is verified	Result
ValidationUtilsTest	Full name	9	valid two/three-word names; single word rejected; null/empty rejected; single-letter second word rejected; extra and leading/trailing whitespace tolerated; whitespace-only rejected	**PASS**
ValidationUtilsTest	Email	8	valid address and subdomain/plus-tag address accepted; missing @, missing domain dot, missing local part, containing a space, bare @, null and empty all rejected	**PASS**
ValidationUtilsTest	Phone	6	exactly 9 digits accepted; 8 or 10 digits rejected; letters rejected; null/empty rejected	**PASS**
ValidationUtilsTest	Password	7	meets all rules accepted; missing uppercase, lowercase, number, special character, or minimum length each individually rejected; null rejected	**PASS**
ValidationUtilsTest	Password matching	3	identical accepted; different rejected; null rejected	**PASS**
ValidationUtilsTest	Email matching	4	identical accepted; different case accepted; different addresses rejected; null rejected	**PASS**
BookingRulesTest	Slot blocks	9	a 60-minute booking at 10:00 reserves four 15-minute blocks; a 45-minute booking at 10:15 reserves three; 30 minutes reserves two; an invalid start time is handled; block IDs use the location or online; overlapping bookings share a block, back-to-back bookings do not, and the same time at different locations does not	**PASS**
BookingRulesTest	Time helpers	3	HH:mm parsed to minutes; invalid input (25:00, 8, ab:cd, null) returns the fallback; minutes formatted back to HH:mm	**PASS**
BookingRulesTest	Open-day text	5	Monday–Friday shown as a range; separate days comma-separated; two consecutive days; range plus single day; no days = "Closed"	**PASS**
BookingRulesTest	Open days and hours in force	4	defaults to Monday–Friday when none saved; saved days respected; hours default to 08:00–18:00; saved hours used	**PASS**
BookingRulesTest	Status handling	3	a missing status is treated as pending; pending/confirmed are active; cancelled/completed are not	**PASS**
WelcomeActivityTest (Espresso)	Welcome screen	1	Register and Login buttons are visible and correctly labelled	**PASS**

4.2 Usability testing

4.2.1 Checklist usability tests (U1 to U12)

ID	Feature	What was tested	Expected result	Actual result	Status	Notes / evidence
U1	Usability	Dropdown fields	Rounded border + visible arrow on all Spinners	As expected.	**PASS**	Evidence: U1
U2	Usability	Password fields	Eye icon toggles visibility correctly on Login and Register	As expected.	**PASS**	Evidence: U2
U3	Usability	Dark Mode	All screens switch palette, remain readable; status badges show bold white text with good contrast	As expected, after the fix for B9.	**PASS**	Evidence: U3. Retest of defect B9.
U4	Usability	Scroll indicators	Visible on screens with content exceeding one view	As expected.	**PASS**	Evidence: U4
U5	Usability	Touch target size	Admin Dashboard's Logout and Settings icons comfortably tappable	As expected.	**PASS**	Evidence: U5
U6	Usability	Register checkbox text	Both consent sentences fully visible; "(optional)" has clear spacing below it	As expected, after the fix for B8.	**PASS**	Evidence: U6. Retest of defect B8.
U7	Usability	Settings long labels	"About", "FAQ" etc. wrap to two lines cleanly, never cut off	As expected.	**PASS**	Evidence: U7
U8	Usability	Logout dialog wording	States that logging out requires logging in again; button order Cancel (left), Log Out (right)	As expected.	**PASS**	Evidence: U8
U9	Usability	Tablet layout — centred content	On a tablet, content is centred with generous margins, not stretched edge-to-edge	As expected, after the fix for B11.	**PASS**	Evidence: U9. Retest of defect B11.
U10	Usability	Tablet layout — no dead space	Fixed-content screens balance vertically on a tall tablet screen	As expected, after the fix for B11.	**PASS**	Evidence: U10. Retest of defect B11.
U11	Usability	Tablet font scaling	Text is visibly larger on tablet than on phone, without layout breaking	As expected.	**PASS**	Evidence: U11
U12	Usability	Landscape orientation	Home, Book Appointment, Register remain usable and scrollable in landscape on phone and tablet	As expected.	**PASS**	Evidence: U12

4.2.2 UX/UI tests (UX-01 to UX-24)

ID	Feature	What was tested	Expected result	Actual result	Status	Notes / evidence
UX-01	Visual consistency	Colour palette across all 22 screens	Same primary colour, background and text colours everywhere; status colours consistent (pending, confirmed, cancelled, completed each has one colour).	As expected.	**PASS**	Evidence: UX-01
UX-02	Visual consistency	Fonts and sizes	Titles, body text and button labels use consistent sizes; no text smaller than readable (about 12sp).	As expected.	**PASS**	Evidence: UX-02
UX-03	Visual consistency	Buttons	Primary buttons look the same on every screen; disabled buttons look disabled (for example Submit while loading).	As expected.	**PASS**	Evidence: UX-03
UX-04	Visual consistency	Spacing and alignment	Equal margins left and right, no overlapping or clipped text, nothing cut off at the bottom.	As expected.	**PASS**	Evidence: UX-04
UX-05	Visual consistency	Dark mode	Switch on in Settings. Every screen stays readable: no dark text on dark background, no white blocks. Setting survives closing and reopening the app.	As expected.	**PASS**	Evidence: UX-05
UX-06	Visual consistency	Screen sizes and rotation	Small phone and tablet both look correct. Rotate on Register, Book Appointment and Admin Dashboard: fields stay filled and nothing crashes.	As expected, after the fix for B12.	**PASS**	Evidence: UX-06. Retest of defect B12.
UX-07	Visual consistency	Touch targets	Buttons and icons are easy to tap with a thumb (about 48dp).	As expected.	**PASS**	Evidence: UX-07
UX-08	Visual consistency	Branding	App name, logo and colours match on Welcome, Login, Home and About.	As expected.	**PASS**	Evidence: UX-08
UX-09	Clarity and ease of use	First-time user test: ask someone who has never seen the app to register, book and cancel without help. Time them and note where they hesitate.	The task is completed without help; hesitation points are recorded.	As expected.	**PASS**	Evidence: UX-09
UX-10	Clarity and ease of use	Labels and hints	Every input has a clear label or hint (phone shows that the +27 prefix is fixed, password requirements are visible).	As expected.	**PASS**	Evidence: UX-10

ID	Feature	What was tested	Expected result	Actual result	Status	Notes / evidence
UX-11	Clarity and ease of use	Booking screen flow	Service, type, location (only when In-Person), duration, date, time, notes: order makes sense, Online shows a note instead of a location.	As expected.	**PASS**	Evidence: UX-11
UX-12	Clarity and ease of use	Empty states: a new client opens My Bookings; the Admin Dashboard with no bookings	A friendly empty message appears on each.	As expected.	**PASS**	Evidence: UX-12
UX-13	Clarity and ease of use	Loading feedback	Progress indicator shows while booking, logging in and loading lists; the user is never left looking at a frozen screen.	As expected.	**PASS**	Evidence: UX-13
UX-14	Clarity and ease of use	Offline banner: turn off data, then turn it back on	A banner appears on supported screens; booking is blocked with a clear message. When data is back on, the banner disappears.	As expected.	**PASS**	Evidence: UX-14
UX-15	Error and confirmation messages	Register with a one-word name	Inline error that two names are needed.	As expected.	**PASS**	Evidence: UX-15
UX-16	Error and confirmation messages	Register with 8 digits in phone	Inline phone error (9 digits required after +27).	As expected.	**PASS**	Evidence: UX-16
UX-17	Error and confirmation messages	Register with mismatched email or password confirmation	Inline mismatch errors.	As expected.	**PASS**	Evidence: UX-17
UX-18	Error and confirmation messages	Register without ticking consent	Red consent error, no account created.	As expected.	**PASS**	Evidence: UX-18
UX-19	Error and confirmation messages	Register with a weak password	The password rules line turns red and bold next to the field (no toast), consistent with the other fields.	As expected.	**PASS**	Evidence: UX-19
UX-20	Error and confirmation messages	Login with empty fields	'This field is required' appears on each empty field.	As expected.	**PASS**	Evidence: UX-20

ID	Feature	What was tested	Expected result	Actual result	Status	Notes / evidence
UX-21	Error and confirmation messages	Login with wrong password	Clear failure message that does not reveal which part was wrong.	As expected.	**PASS**	Evidence: UX-21
UX-22	Error and confirmation messages	Booking on a closed day, a public holiday, and outside hours	Specific messages naming the open days, the holiday, or the opening and closing times.	As expected.	**PASS**	Evidence: UX-22
UX-23	Error and confirmation messages	Booking on a taken time	Clear message that the time is already reserved.	As expected.	**PASS**	Evidence: UX-23
UX-24	Error and confirmation messages	Successful booking, cancel, decline, delete, paid toggle	A confirmation message appears for each, and destructive actions (cancel, decline, delete) show a confirmation dialog first.	As expected.	**PASS**	Evidence: UX-24

4.3 Edge-case and invalid-input testing

4.3.1 Edge cases (E1 to E8)

ID	Feature	What was tested	Expected result	Actual result	Status	Notes / evidence
E1	Edge case	Submit booking form with fields missing	Specific message per missing field	As expected.	**PASS**	Evidence: E1
E2	Edge case	Attempt booking while offline	Clear offline message, no crash	As expected.	**PASS**	Evidence: E2
E3	Edge case	Attempt login while offline	Clear offline message, no crash	As expected.	**PASS**	Evidence: E3
E4	Edge case	Open My Bookings offline, never loaded before	Empty state shown, no crash	As expected.	**PASS**	Evidence: E4
E5	Edge case	Holiday API unreachable	Booking still works on open-day and hours rules; toast notes holidays couldn't be checked	As expected.	**PASS**	Evidence: E5
E6	Edge case	Search with no matching results	Empty state shown, no crash	As expected.	**PASS**	Evidence: E6
E7	Edge case	Export CSV with zero bookings in the filtered list	Export still runs with a header row only	As expected.	**PASS**	Evidence: E7
E8	Edge case	Back button on a non-root screen (e.g. Book Appointment)	Normal back navigation — the "press again to exit" behaviour applies only to Home/Admin Dashboard	As expected.	**PASS**	Evidence: E8

4.3.2 Invalid input and failure scenarios across the whole test set

These tests, recorded in other sections, also deliberately used blank, invalid, boundary, offline or hostile input. The app either prevented the action or showed a clear message in every case.

Scenario	Test IDs	Result
Blank forms and empty fields (registration, login, booking)	UX-15, UX-20, F2, E1, BK-14, PF5	Prevented; field-level messages
Invalid name, phone, email, password, mismatched confirmation, missing consent	UX-15 to UX-19, F2, F24, DB-06, DB-07, PF3, PF4	Rejected with inline errors
Wrong credentials, unverified email, duplicate email	DB-02, DB-04, DB-05, UX-21, F3	Blocked with clear message; no session or duplicate created
Closed day, public holiday, outside hours, off quarter-hour times	UX-22, BK-09 to BK-13, PC1, PC2, PC6, PC8, F8	Rejected with a specific reason
Overlapping or taken times, own overlapping booking	UX-23, BK-02, BK-03, BK-07, PC3, PC4, F7, INT-10	Rejected; no partial data saved
Invalid practice hours or no open days	DB-20, PC7	Rejected; nothing saved
No internet: login, booking, cancel, registration, app restart	E2 to E5, BK-21, DB-12, PE1, PE3, NAV-18, INT-07, INT-08	Clear offline message; no crash; no orphaned data
Holiday API unreachable	E5, INT-05	Booking still checks other rules; user informed
Abnormal data (booking with missing status)	PE2	No crash; shown as Pending
Search with no results, export with no bookings	E6, E7	Empty state / header-only file; no crash
Double tap, rotation, back button on any screen	NAV-14 to NAV-16, E8, U12, UX-06	One action only; state kept; normal back
Forbidden database actions (read others' data, role escalation, tampering with bookings and slots)	PM-01, PM-04, PM-06, PM-07, PM-09, PM-10, PM-12, PM-13, PM-15 to PM-18, PA2	Denied by security rules

5. Bugs found and fixed

Defects were found during development (B1 to B12) and during the final testing audit (B13 to B27). Critical defects (security, crashes, data integrity) were fixed first. The retest column names the tests that were repeated after each fix.

No.	Type	Severity	Defect found	Root cause	Steps taken to resolve	Retest	Status
B1	Build	Medium	Admin Dashboard XML failed to compile after adding app:tint	Missing xmlns:app namespace on the root layout tag	Added the namespace declaration	Clean build	**FIXED**
B2	UI	Medium	Logout unreachable on Settings	Buttons sat below the visible scroll area, no scroll indicator	Added a scroll thumb and repositioned logout	F19, F16	**FIXED**
B3	Environment	Low	Emulator "Unknown error" on run	Emulator process had hung	Cold-booted the emulator	Emulator restarted and app run	**FIXED**
B4	Build	Low	PackageInfo import error	Wrong package path (android.pm vs android.content.pm)	Corrected the import	Clean build	**FIXED**
B5	Build	Low	Test files could not resolve JUnit/Espresso imports	Files placed in src/main/java instead of src/test/src/androidTest	Moved to the correct source sets	Unit tests run	**FIXED**
B6	Build	Medium	strings.xml failed to parse	A stray invalid closing tag was accidentally included	Removed the invalid tag	Clean build	**FIXED**
B7	Crash	High	Admin login crashed after the date-filter redesign	tvFilterDate changed from TextView to LinearLayout in XML but the Java field type wasn't updated	Changed the field type to View	Admin login retested	**FIXED**
B8	UI	Low	Register consent text clipped	Checkbox row height/text-wrap not properly configured	Rows set to wrap_content, maxLines/ellipsize="none", explicit bottom margins	U6	**FIXED**
B9	UI	Low	Status badges hard to read in light mode	Black text on dark badge backgrounds	Bold white badge text for all four statuses	U3	**FIXED**
B10	Navigation	Medium	Back on Home returned to Welcome (looked like a logout)	Welcome stayed on the back stack beneath Home; FLAG_ACTIVITY_CLEAR_TOP did not clear it	Double-press-to-exit on Home/Admin Dashboard; navigation flags changed to NEW_TASK \	F21, NAV-02, NAV-14	**FIXED**
B11	UI	Low	Register did not centre vertically on tablet	NestedScrollView + fillViewport + weight-based centring does not reliably stretch	Switched to a plain ScrollView with a constrained content column	U9, U10	**FIXED**

No.	Type	Severity	Defect found	Root cause	Steps taken to resolve	Retest	Status
B12	UI	Low	Register header was dragged to the middle on tablet	Header and form shared one centred container	Layout restructured so the header stays at the top	U9, U10	**FIXED**
B13	Security	High	A user could register themselves as an admin	The users create rule only checked ownership, not the role value	Rule now requires role == "client" on create	PA1, PM-06	**FIXED**
B14	Security	High	A client could confirm their own booking or mark it paid	The bookings update rule allowed the owner to change any field	Clients limited to status → "cancelled"; everything else admin-only	PA2, PM-12	**FIXED**
B15	Security	High	Any user could overwrite or delete any time slot	bookedSlots allowed any signed-in user to write	Slots carry clientId; create/delete limited to the owner or an admin; update denied	PM-15 to PM-17	**FIXED**
B16	Data integrity	High	Cancel/decline could leave a stuck slot	Status update and slot delete were two separate calls	One Firestore transaction (BookingActions)	PB1, PB2, BK-15	**FIXED**
B17	Functionality	Medium	Bookings had no true Delete	Cancel only set status = "cancelled"	Admin Delete Booking (permanent, confirmed, atomic)	PB3, BK-19	**FIXED**
B18	Logic	High	Two different clients could book overlapping times	Overlap was only checked against the same client's bookings; slot IDs ignored duration	Per-15-minute block reservation inside a transaction	PC3, BK-03	**FIXED**
B19	Logic	Medium	An appointment could finish after closing time	Only the start hour was compared with closing time	Start + duration checked against closing; re-checked on duration change and submit	PC1, BK-11	**FIXED**
B20	Logic	Medium	Admin's Operating Days setting had no effect	The booking screen hardcoded Saturday/Sunday and ignored the saved value	Structured openDays list edited with checkboxes and enforced when booking	PC6, PC7	**FIXED**
B21	Functionality	Medium	The Booking Reminders switch did nothing	The preference was saved but never read before notifying	Enforced in NotificationHelper; reminders cancelled/rescheduled on toggle	PD1 to PD5	**FIXED**
B22	Data integrity	Medium	A failed registration could leave an orphaned login	Auth account created, then the profile write failed with no rollback	Auth account deleted if the profile write fails	PE1, DB-12	**FIXED**
B23	Crash	High	A booking with no status could crash screens	Unguarded switch on a null status	StatusUtils treats missing status as pending	PE2	**FIXED**
B24	Logic	Medium	Holidays only checked for the current year	Holidays fetched once, for the current year	Fetched and cached per year	PC8	**FIXED**

No.	Type	Severity	Defect found	Root cause	Steps taken to resolve	Retest	Status
B25	Validation	Low	Email validation accepted @ and a@b	Rule was contains("@")	Regex requiring local part, @ and a dotted domain; tests updated	PF3	**FIXED**
B26	Code quality	Low	Unused default test stubs and .idea/ committed	Never removed	Deleted; .idea untracked and ignored	PF1, PF9	**FIXED**
B27	UI	Low	Opening Settings could show a false "reminders enabled" toast	The switch listener was attached before the saved value loaded	Listener attached only after the saved value is applied	PD3	**FIXED**

5.1 Most significant defects in detail

Defect	How it was found	Why it mattered	Fix and verification
B13 User could register as admin	Final implementation audit of the security rules	Anyone could gain full admin access to every booking and client record	Create rule now requires role == client. Retested: PM-06 denied, PA1 normal registration still works
B14 Client could confirm or pay for own booking	Final implementation audit of the security rules	Clients could bypass the admin approval and payment process	Client updates limited to status cancelled only. Retested: PM-12 denied, PA2 denied, PA3 full flow still works
B15 Any user could change any time slot	Final implementation audit of the security rules	A user could delete other clients' reservations and cause double bookings	Slots carry clientId; only the owner or admin may create or delete; updates denied. Retested: PM-15 to PM-17
B16 Cancel or decline could leave a stuck slot	Final implementation audit: status update and slot delete were separate calls	A time could stay blocked forever	One Firestore transaction updates the booking and frees the slots. Retested: PB1, PB2, PC5, BK-15
B18 Overlapping bookings by different clients	Final audit and BK-03: overlap check only covered the same client's bookings	Two clients could be booked at the same time	Every 15-minute block of the appointment is reserved in one transaction. Retested: PC3, BK-03, BK-08
B23 Missing booking status crashed screens	Final audit; PE2 deletes the status field in the Console	The admin dashboard crashed	StatusUtils treats a missing status as pending. Retested: PE2, no crash

6. Stability: does the system run without crashing?

The final version of the app was used end to end on a real phone and a real tablet, as a client and as an administrator, without a crash. Specific stability checks:

- Abnormal data: a booking with its status field deleted does not crash the dashboard or the detail screen (PE2).
- Abnormal conditions: opening the app offline, logging in or booking offline, the holiday API being unreachable, and a failed registration all give clear messages and do not crash (PE1, PE3, E2 to E5, BK-21).
- Rotation, rapid double taps and the back button on every major screen cause no crash, no duplicate action and no lost data (NAV-14 to NAV-16, E8, U12).
- Two crash defects were found and fixed during development and testing (B7 and B23), and both were retested.
- The 61 automated tests and the Espresso test pass on every run, and the project builds cleanly from a fresh checkout (PF1, PF2).
- Firebase Crashlytics is integrated in the app so any crash on a device is recorded for the developers.

7. Limitations and suggestions for improvement

- Clients cannot edit or reschedule a booking; they cancel and book again. A reschedule transaction that releases the old time blocks and claims the new ones would improve this.
- All Online appointments share one calendar, so only one Online appointment can exist at a given time (BK-22). This suits one online practitioner but should be revisited if more are added.
- Notifications are local; instant push notifications to a client when an admin acts need a server-side function.
- Account deletion is requested in the app and completed by the administrator.
- Test coverage of the Android activities is mostly manual. Adding more Espresso tests for the booking and login flows would let them run automatically.

8. Conclusion

The MMM Booking App was tested systematically across functional, usability, edge-case and security areas using 258 recorded tests plus an Espresso UI test. Every test passed after defects were fixed, 27 defects were found, resolved and retested, including three serious security holes and two crashes, and the final app runs without crashing on a real phone and tablet. The remaining limitations are documented in section 7.

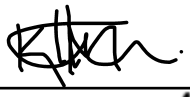
Appendix A: Evidence index

Each test ID has a matching screenshot or screen recording in Testing_Evidence_MMM_Booking_App-Group11.pdf, in the repository root. The first page of that document records the device models and Android versions used.

Test ID prefix	Test group	Type of evidence	Pages in evidence PDF
UT1, UT2	Automated unit tests and Espresso test	Screenshot of 61 tests passing in Android Studio; Espresso result	8 to 9
PA to PF	Hardening verification	App screenshots, Rules Playground results, Firebase Console, adb output (PD4), Logcat (PE5)	10 to 29
F1 to F25	Manual functional tests	App screenshots and short recordings	30 to 42
U1 to U12	Usability tests	Phone and tablet screenshots	43 to 47
E1 to E8	Edge-case tests	Offline, empty-state and error screenshots	48 to 51
UX-01 to UX-24	UX/UI tests (Faith Murau)	Screens in light and dark mode, message screenshots	52 to 63
NAV-01 to NAV-18	Screens and navigation (Khauhelo Moqechane)	Screen inventory and navigation screenshots	64 to 73
DB-01 to DB-21	Authentication and data (Ignatius Mayibongwe Khumalo)	App screenshots with Firebase Console documents	74 to 84
BK-01 to BK-22	Booking logic (Ignatius Mayibongwe Khumalo)	Two-device screenshots, Console slot documents	85 to 95
PM-01 to PM-18	Access control (Ignatius Mayibongwe Khumalo)	Rules Playground results (allowed and denied)	96 to 104
INT-01 to INT-12	Integration (all three members)	Two-phone screenshots and Console screenshots	105 to 111
B1 to B27	Defect log	Defect, root cause and resolution table	112 to 113

9. Sign-off

We confirm that the tests above were run on the devices listed on the first page and that the screenshots are genuine results from those runs.

Student Signature(s)	Date
Ignatius Mayibongwe Khumalo	
Moqechane, Khauhelo	
Faith Murau	